

DIGITAL MODULATION TECHNIQUES ^{DM}

Basically there are two ways of transmission of digital signals are baseband and passband transmission

1. Baseband Transmission: The digital data generated by a digital source are transmitted directly through the transmission channel without any modulation is called baseband transmission.

2. Passband Transmission: The digital data are generated by a digital source is modulated with a high frequency carrier signal and ~~then~~ transmitting through a transmission channel is called passband transmission

The baseband transmission is preferred for short distance transmission where as passband transmission is preferred for long distance transmission.

Digital modulation is defined as shifting the parameter of carrier wave in accordance with the digital data signal is called digital modulation. There are number of digital modulation formats, which are listed below

- * ASK
 - * PSK
 - * FSK
 - * DPSK
 - * QPSK
 - * M-ary PSK
 - * M-ary FSK
 - * QASK or QAM
 - * MSK
- } Basic Modulation technique
- } Enhanced Modulation technique
- } Special Modulation Technique

TYPES OF RECEPTION: There are two types of detections

1. Coherent (synchronous) detection:- In this type a carrier signal is locally generated at the receiver, which is exactly phase locked with the carrier signal used at the transmitter. Hence it is called coherent or synchronous detection.

2. Non-coherent detection:- In this type, the carrier signal need not to be generated at the receiver, instead the signal is pass through band-pass filter followed by a Envelop detector for detecting baseband signal. Such type of detection is called Non-coherent type of detection.

REQUIREMENTS OF DIGITAL COMMUNICATION SYSTEM.

If we can design any digital communication system, the designed system has to meet following requirements

- * Maximum data transmission rate
- * Minimum probability of symbol error or bit error
- * Should require minimum transmitting power
- * minimum channel Bandwidth
- * Maximum resistance to interfering signals
- * Minimum circuit complexity

ADVANTAGES

- * Long distance transmission is possible
- * Analog channels can also be used for transmission
- * Multiplexing technique can be used for bandwidth Conservation
- * Probability of ISI and cross talk are very minimum
- * The digital modulated signal can also be transmitted through a wireless channel

AMPLITUDE SHIFT KEYING [ASK]

DM-2

ASK is a simplest form of digital modulation technique, in which the amplitude of carrier is shifted or keyed in accordance with incoming data signal. The ASK is also called ON-OFF Keying (OOK), because either a carrier signal is present for the symbol duration or absent for the symbol duration depending upon whether symbol '1' or symbol '0' being transmitting.

⇒ Generation of ASK

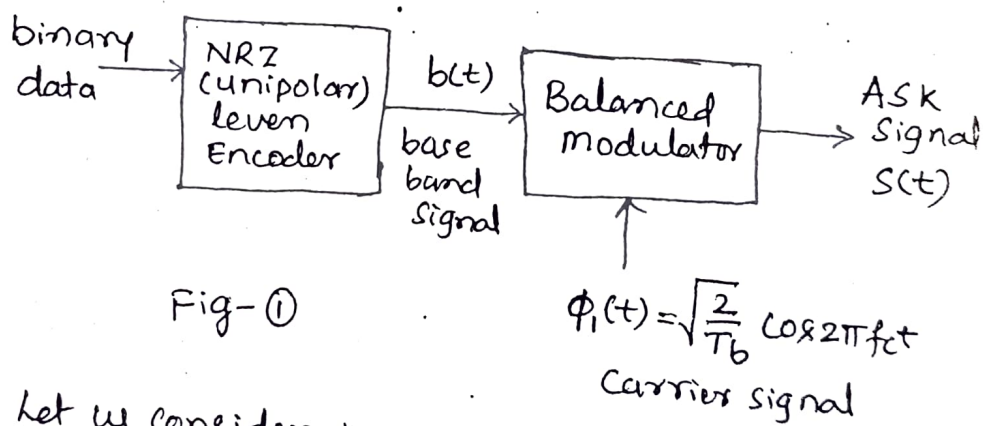


Fig-①

Let us consider the carrier signal $\phi_1(t) = \sqrt{\frac{2}{T_b}} \cos 2\pi f_c t$ and $b(t)$ is a base band signal and ASK modulated then the resulting signal $S(t)$ is given by

$$S(t) = \begin{cases} \sqrt{2P_s} \cos 2\pi f_c t & 0 \leq t \leq T_b \text{ for } s_y = 1 \\ 0 & 0 \leq t \leq T_b \text{ for } s_y = 0 \end{cases} \quad \text{--- ①}$$

From equation ① the signal $S(t)$ is given by

$$S(t) = \sqrt{\frac{2P_s T_b}{T_b}} \cos 2\pi f_c t$$

$$S(t) = \sqrt{P_s T_b} \sqrt{\frac{2}{T_b}} \cos 2\pi f_c t \quad \text{--- (2)}$$

$$\therefore S(t) = \sqrt{P_s T_b} \phi_1(t)$$

Where $P_s T_b = E_b$, energy of a bit

$$\therefore S(t) = \sqrt{E_b} \phi_1(t) \quad \text{--- (3)}$$

The wave form for $b(t)$, $c(t)$ and $s(t)$ are shown in fig-2

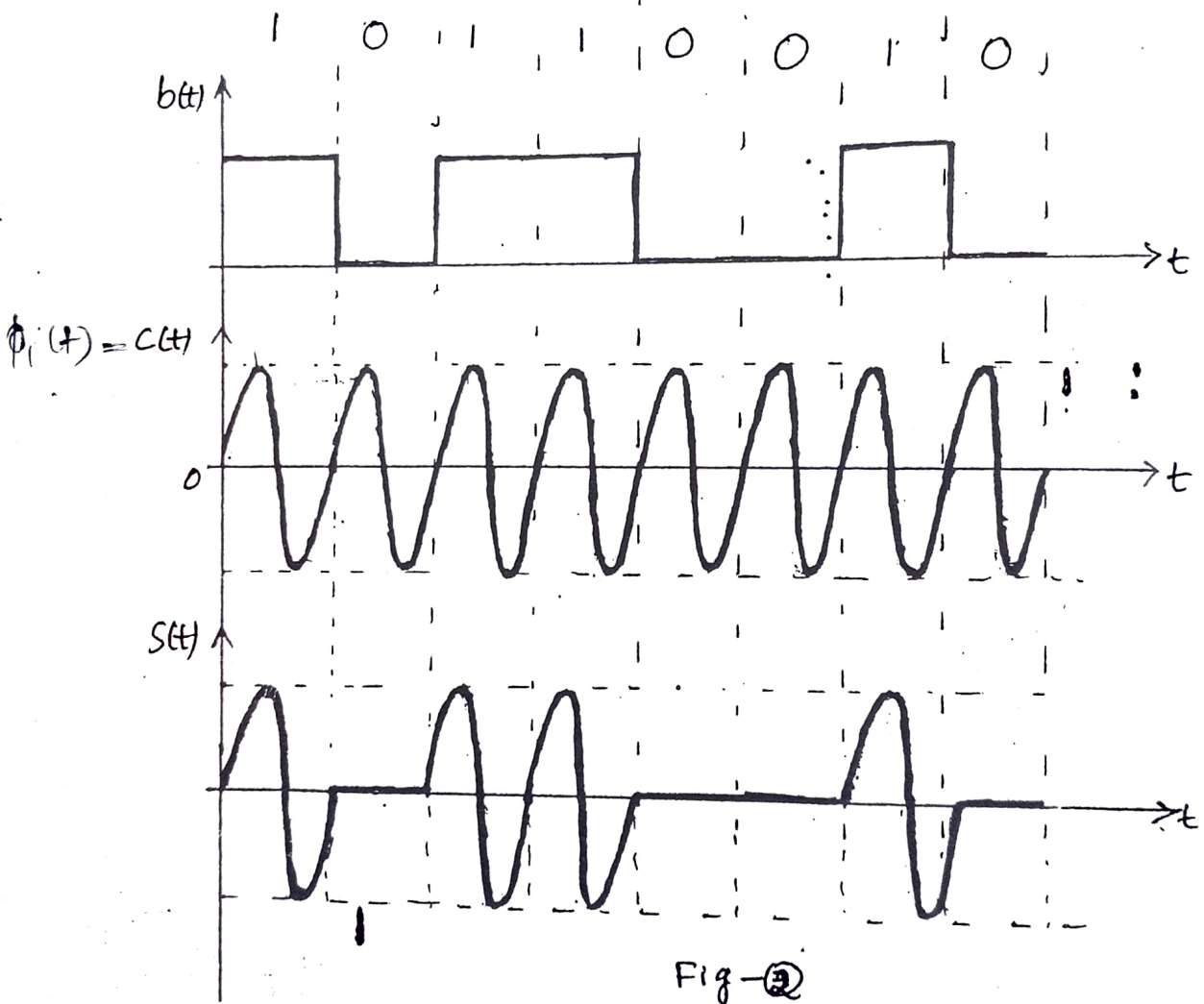


Fig-2

$\Rightarrow$ ASK Detection:- The block diagram of coherent types of ASK detection is shown in fig-3. The ASK modulated signal $s(t)$ taken from the output of channel is applied to correlator consisting of multiplier and integrator. The locally generated carrier is applied to the multiplier shown in fig-3

DM-3

The output of multiplier is integrated over one-bit interval, since integrator behaves like a low-pass filter the high frequency carrier signal present at the output of multiplier is suppressed and only data signal waveform is allowed. The waveform coming out from integrator is sampled for every bit (T_b) interval and the resulting sample is compared with the threshold level of decision device and makes the decision either in favour of Symbol - '1' or Symbol - '0'. This type of detection is called Coherent detection.

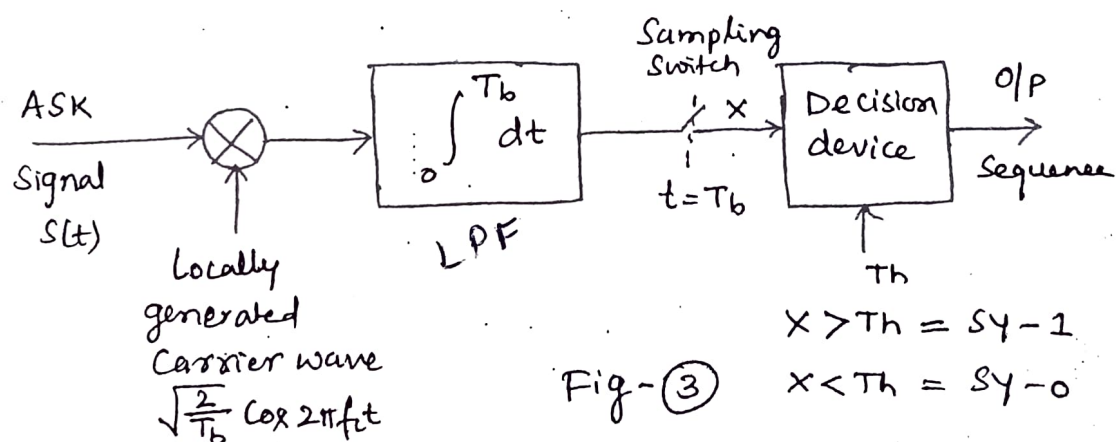


Fig - (3)

In case of Non-coherent detection the carrier signal is not locally generated at the receiver, instead the received signal $S(t)$ is directly pass through a band-pass filter with a center frequency of f_c , then it is pass through a envelop detector to get base band signal. Agein the output of envelop detector is sampled and these samples are applied to decision device to

make decision either in favour of symbol -1 or in favour of symbol -0'. The block diagram of Non-Coherent detection is shown in fig - (4)

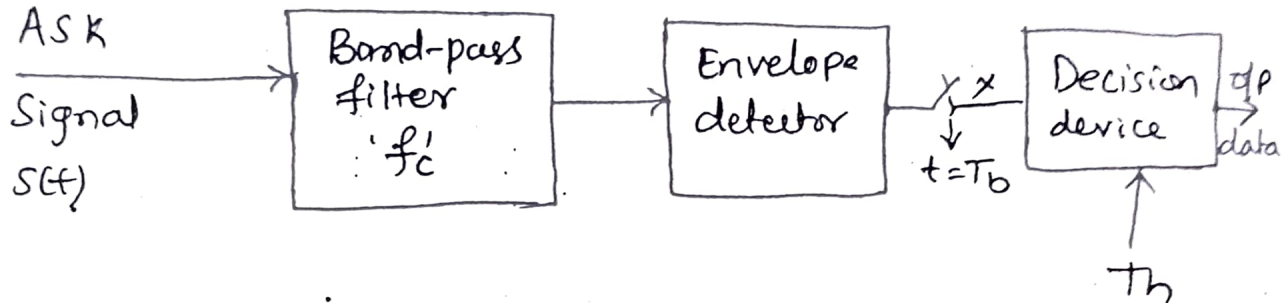


Fig - (4).

$$x > T_h = S_y - 1$$

$$x < T_h = S_y - 0$$

⇒ SPECTRUM OF ASK SIGNAL

DM-1

When any baseband signal is modulated with a high frequency carrier the resulting signal $s(t)$ is called a band-pass signal. The power spectral density of any band-pass signal is given by

$$S_s(f) = \frac{1}{4} [S_B(f-f_c) + S_B(f+f_c)] \quad \text{--- (4)}$$

where $S_B(f) \Rightarrow$ Spectrum of baseband signal.

Since the incoming data's are represented by an NRZ unipolar format the power spectral density of it is given by

$$S_B(f) = \frac{a^2}{4} \delta(f) + \frac{a^2 T_b}{4} \text{sinc}^2(f T_b) \quad \text{--- (5)}$$

Where $T_b \Rightarrow$ bit duration or symbol duration

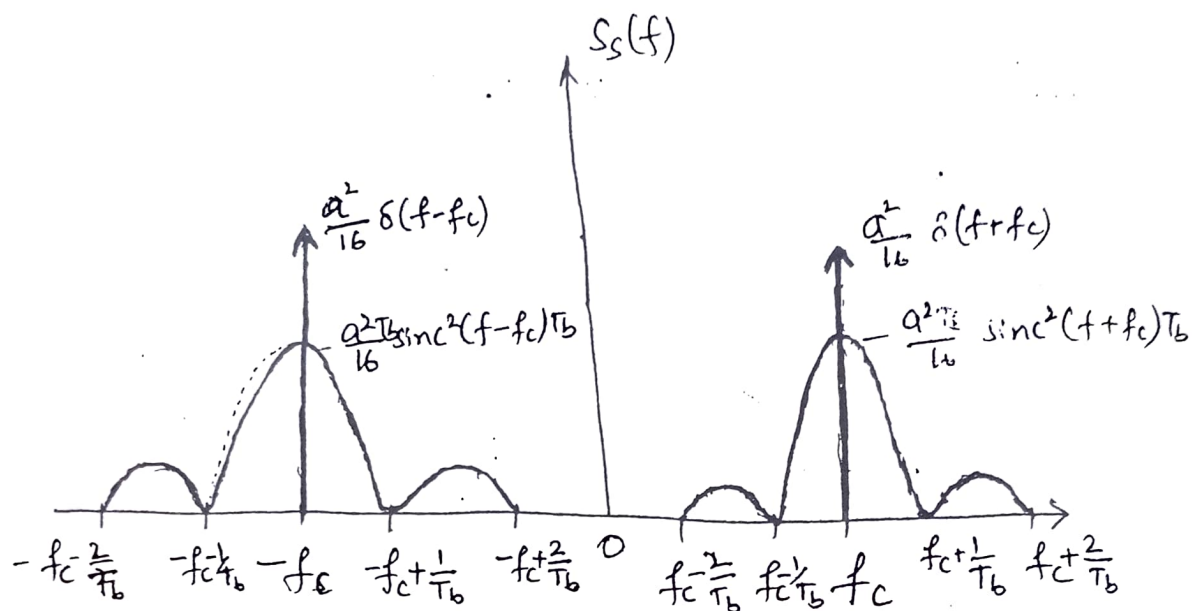
$a \Rightarrow$ product of amplitude of carrier and baseband signal.

$$\therefore a = \sqrt{\frac{2}{T_b}} \times \sqrt{E_b} = \sqrt{\frac{2E_b}{T_b}} \quad \text{--- (6)}$$

Substitute equation (5) in (4) we have

$$\begin{aligned} S_s(f) = & \frac{a^2}{16} \delta(f-f_c) + \frac{a^2 T_b}{16} \text{sinc}^2[(f-f_c) T_b] \\ & + \frac{a^2}{16} \delta(f+f_c) + \frac{a^2 T_b}{16} \text{sinc}^2[(f+f_c) T_b] \end{aligned} \quad \text{--- (7)}$$

From equation (7), we can able to draw a PSD for Signal $s(t)$ i.e. $S_s(f)$ is shown in fig - (6)



From the Spectrum of ASK signal, the bandwidth required for ASK signal is given by

$$\begin{aligned}
 BW &= \left[f_c + \frac{1}{T_b} \right] - \left[f_c - \frac{1}{T_b} \right] \\
 &= f_c + \frac{1}{T_b} - f_c + \frac{1}{T_b} \\
 &= \frac{2}{T_b} = \underline{\underline{2f_b}}
 \end{aligned}$$

The bandwidth required for ASK signal is twice the bit rate of the base band signal.

Note P_e for ASK is same as P_e in PCM

PHASE SHIFT KEYING [PSK]

The PSK is defined as shifting or keying phase of the carrier in accordance with baseband signal.

Again PSK belongs to basic digital modulation technique consists of two symbols are '1' and '0'. In PSK for every change in symbol the carrier signal is phase shifted by 180° , hence PSK modulated signal is defined as

$$S(t) = \begin{cases} \sqrt{\frac{2E_b}{T_b}} \cos 2\pi f_c t & 0 \leq t \leq T_b \text{ for } s_y = 1 \\ -\sqrt{\frac{2E_b}{T_b}} \cos 2\pi f_c t & 0 \leq t \leq T_b \text{ for } s_y = 0 \end{cases} \quad \text{--- (1)}$$

Correspondingly signal $S(t)$ is separated as $S_1(t)$ & $S_2(t)$

$$\therefore \begin{cases} S_1(t) = \sqrt{\frac{2E_b}{T_b}} \cos 2\pi f_c t & \text{for } s_y = 1 \\ S_2(t) = \sqrt{\frac{2E_b}{T_b}} \cos (2\pi f_c t + \pi) & \text{for } s_y = 0 \end{cases} \quad \text{--- (2)}$$

$\therefore S_1(t)$ & $S_2(t)$ are written as

$$S_1(t) = \sqrt{E_b} \sqrt{\frac{2}{T_b}} \cos 2\pi f_c t$$

$$S_2(t) = -\sqrt{E_b} \sqrt{\frac{2}{T_b}} \cos 2\pi f_c t$$

$$\text{where } \phi_1(t) = \sqrt{\frac{2}{T_b}} \cos 2\pi f_c t \quad \text{--- (3)}$$

$$\therefore \begin{cases} S_1(t) = \sqrt{E_b} \phi_1(t) \\ S_2(t) = -\sqrt{E_b} \phi_1(t) \end{cases} \quad \text{--- (4)}$$

Also signal $s(t)$ is expressed as

$$s(t) = b(t) \sqrt{\frac{2E_b}{T_b}} \cos 2\pi f_c t \quad \text{--- (5)}$$

$$\text{where } b(t) = \begin{cases} +1 & \text{for } s_y = 1 \\ -1 & \text{for } s_y = 0 \end{cases} \quad \text{--- (6)}$$

⇒ Generation of PSK signal

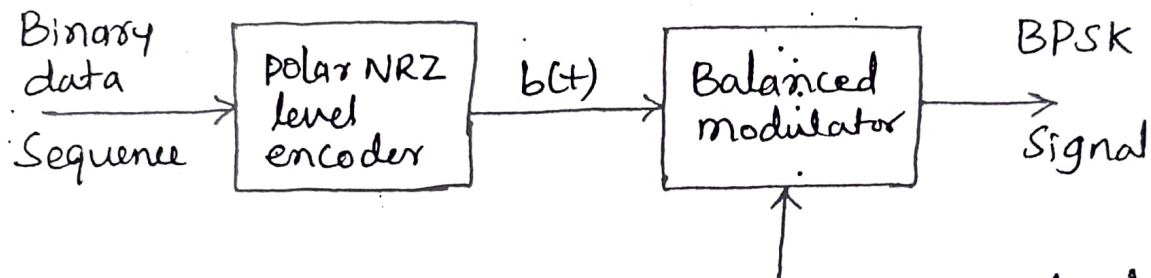


fig - ①

$$\phi_c(t) = \sqrt{\frac{2}{T_b}} \cos 2\pi f_c t$$

Carrier signal.

The block diagram for generation of PSK signal is shown in fig ①. Here the incoming data are encoded by polar NRZ waveform. That is positive amplitude for symbol-1 and negative amplitude for symbol-0 $[b(t)]$. $b(t)$ is modulated with a carrier signal $\phi_c(t)$ and resulting PSK modulated signal is coming out from balanced modulator. The waveform representing $b(t)$, $\phi_c(t)$ and modulated signal $s(t)$ are shown in fig - ②

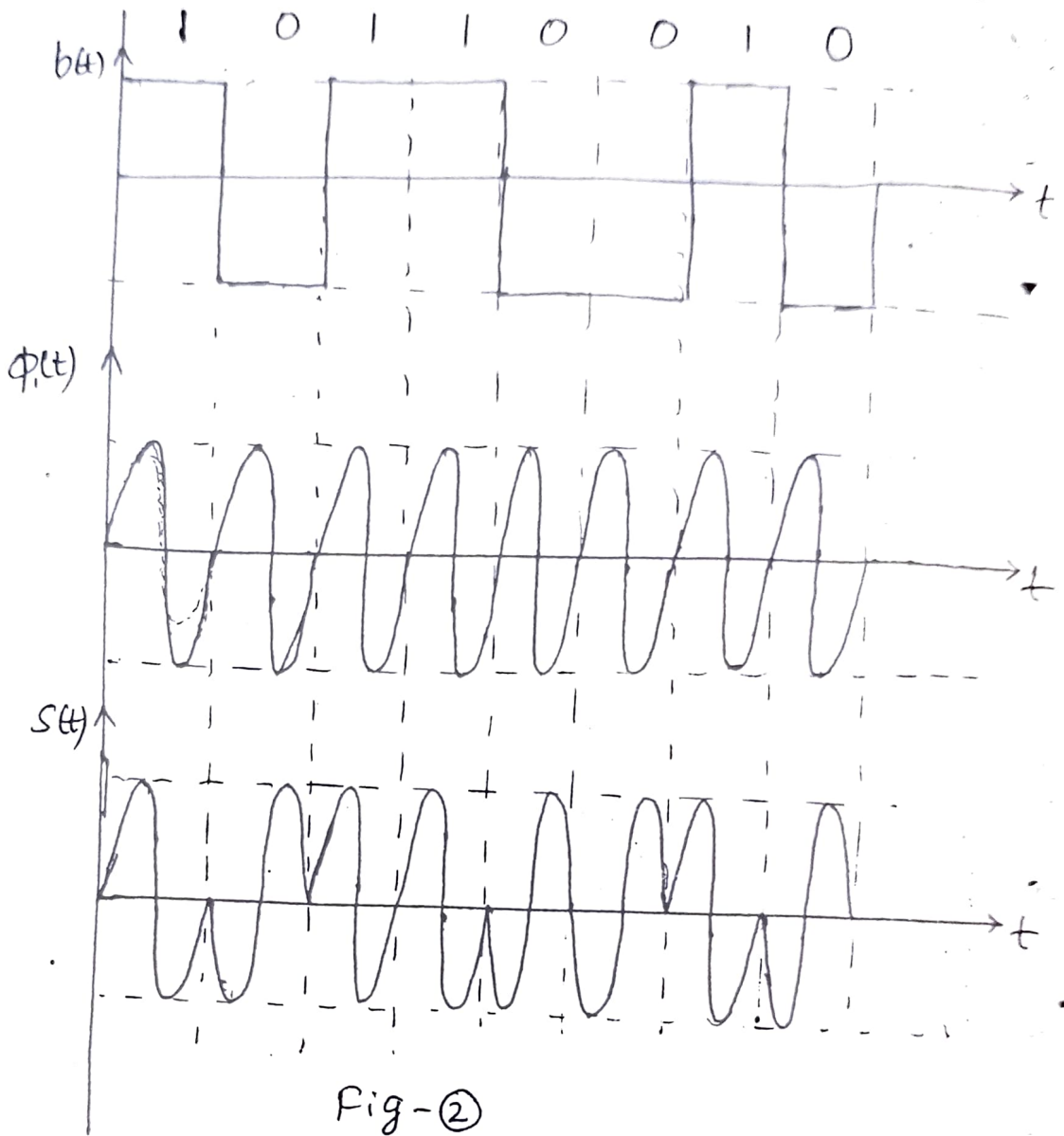


Fig-②

⇒ Detection of BPSK Signal

The block diagrams shown in fig-③ and fig-④ are coherent detection of BPSK signal. Fig-③ is the general block diagram of coherent BPSK detection and fig-④ shows the simplified or locally generated carrier signal at the receiver followed by coherent BPSK detection.

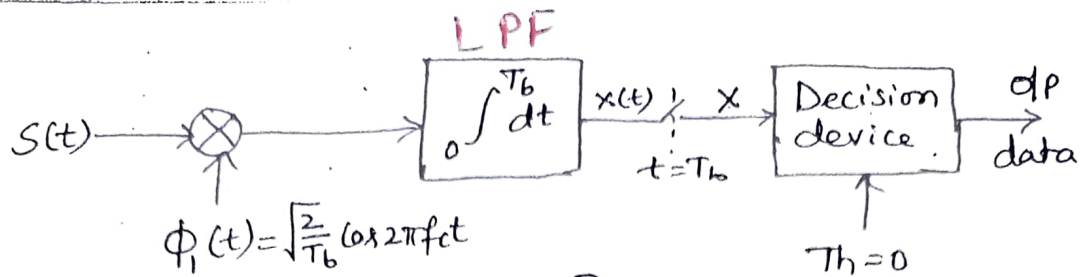


Fig-③

OR

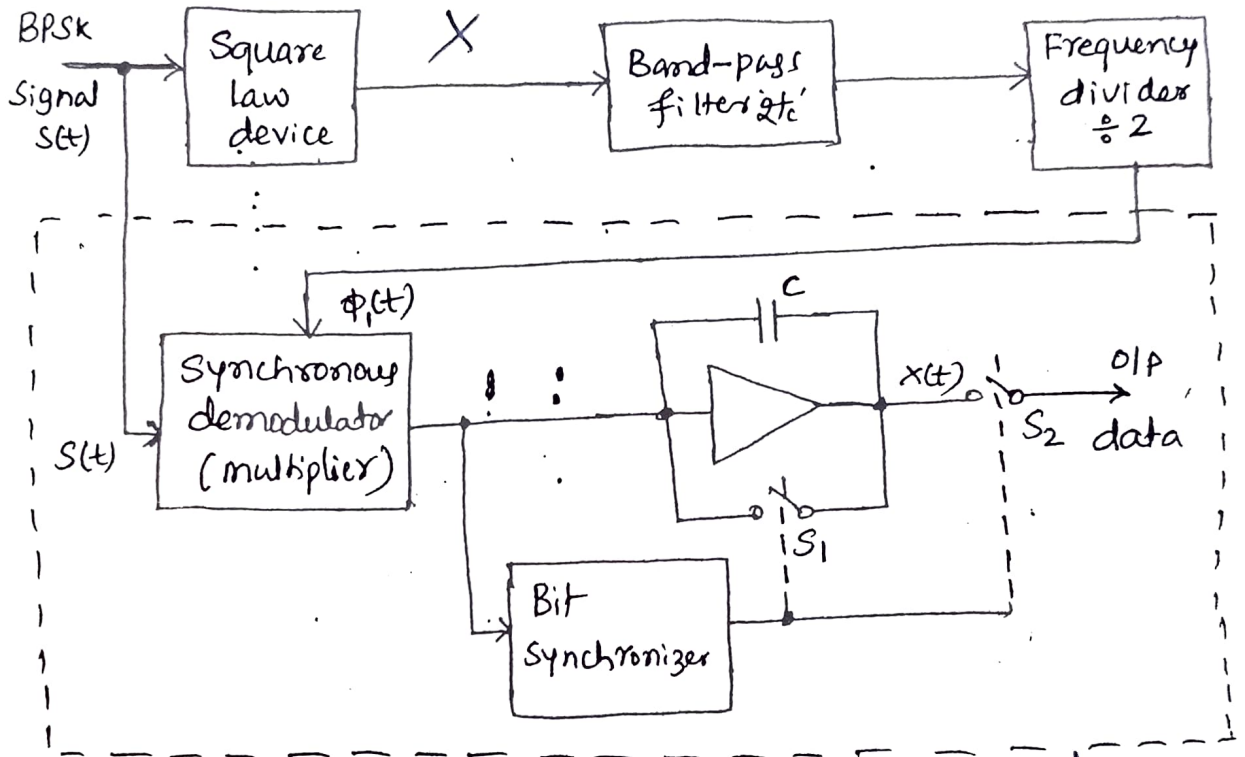


Fig-④

not required

Note: The dotted box of fig-④ is equivalent to fig-③. The outside the dotted box of fig-④ represents locally generation of carrier signal.

* In fig-④, the received signal $s(t)$ is passed through a square law device or signal squarer, the output of it is given by

$$s^2(t) = \cos^2(2\pi f_c t + \theta) \quad \text{--- ⑦}$$

The equation (7) represents the output of square law device in which we have neglected the amplitude of modulated signal $s(t)$. ' θ ' represents the transmission delay

$$\begin{aligned}
 S^2(t) &= \cos^2(2\pi f_c t + \theta) \\
 &= \frac{1 + \cos 2(2\pi f_c t + \theta)}{2} \\
 &= \underbrace{\frac{1}{2}}_{\text{DC Component}} + \underbrace{\frac{1}{2} \cos 2(2\pi f_c t + \theta)}_{\text{AC Component}} \quad \text{--- (8)}
 \end{aligned}$$

(*) The output of square law device is given band pass filter with a center frequency of $2f_c$. The band pass filter eliminates the DC component and allows only AC component. Therefore output of band-pass filter is given by

$$\cos 2(2\pi f_c t + \theta) \quad \text{--- (9)}$$

(*) Again output of band pass filter is pass through a frequency divider, which divides the frequency component by $\frac{2}{2}$, the resulting signal is given by

$$\cos(2\pi f_c t + \theta) \quad \text{--- (10)}$$

by assuming the value of $\theta = 0$, then the signal locally generated at the receiver is exactly same as

the carrier used at the transmitter.

(*) The output of synchronous demodulator or multiplier

$$\begin{aligned} \text{ie } & b(t) \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t + \theta) \times \cos(2\pi f_c t + \theta) \\ &= b(t) \sqrt{\frac{2E_b}{T_b}} \cos^2(2\pi f_c t + \theta) \\ &= b(t) \sqrt{\frac{2E_b}{T_b}} \left[\frac{1 + \cos 2(2\pi f_c t + \theta)}{2} \right] \\ &= b(t) \sqrt{\frac{E_b}{2T_b}} + b(t) \sqrt{\frac{E_b}{2T_b}} \cos 2(2\pi f_c t + \theta) \quad \text{--- (11)} \end{aligned}$$

(*) When equation (11) pass through integrator, only the first term is allowed and 2nd term is suppressed, hence the output of it is given by

$$= b(t) \sqrt{\frac{E_b}{2T_b}} \quad \text{--- (12)}$$

& the recovered sequence is directly proportion to the given input sequence

(*) The bit synchronizer operates the two switches S_1 & S_2
When S_2 closes the sample is given to DD
When S_1 closes the integrator resets.

$\Rightarrow$ Spectrum of BPSK Signal

As we know that the spectrum of bandpass signal is

$$S_s(f) = \frac{1}{4} [S_B(f-f_c) + S_B(f+f_c)] \text{ --- (14)}$$

In PSK modulation, the data signal is represented as NRZ polar and its power spectral density (PSD) is given by

$$S_B(f) = a^2 T_b \text{sinc}^2(f T_b) \text{ --- (15)}$$

Substitute equation (15) in (14) i.e

$$S_s(f) = \frac{a^2 T_b}{4} \text{sinc}^2[(f-f_c) T_b] + \frac{a^2 T_b}{4} \text{sinc}^2[(f+f_c) T_b] \text{ --- (16)}$$

The equation (16) represents the power spectral density of PSK modulated signal. The spectrum of it is shown in fig-(b)

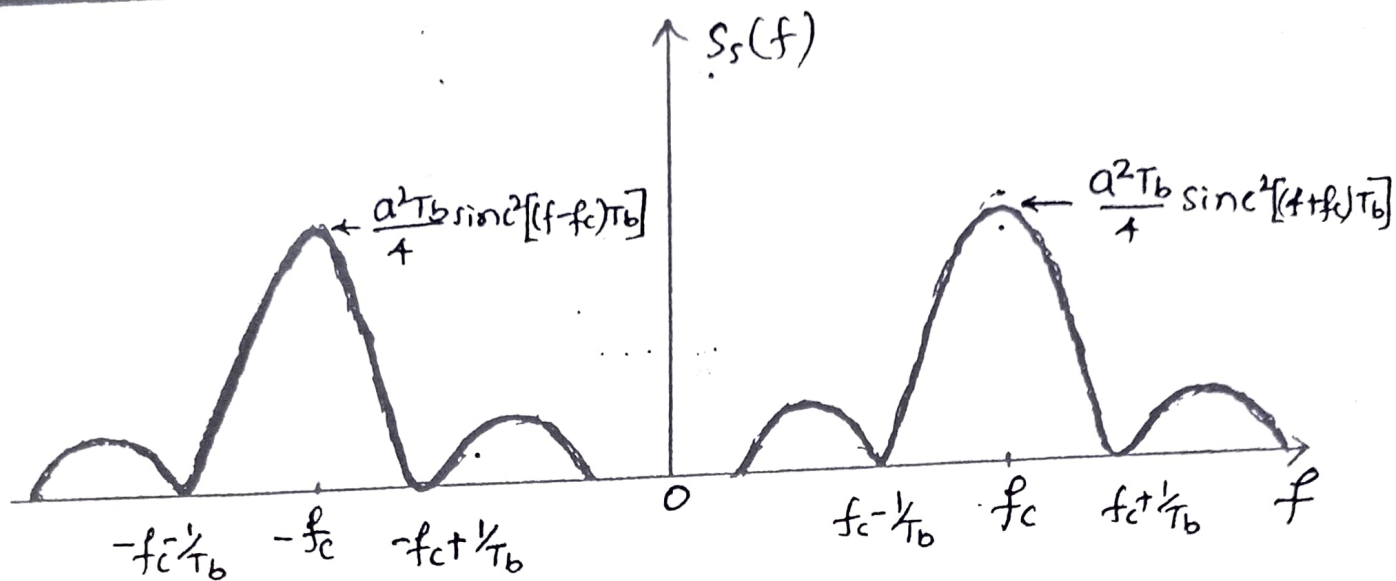


Fig-(6)

⇒ Bandwidth of BPSK signal

From fig-(6) the bandwidth of BPSK signal is

$$BW = \left[f_c + \frac{1}{T_b} \right] - \left[f_c - \frac{1}{T_b} \right]$$

$$= \left[f_c + f_b \right] - \left[f_c - f_b \right]$$

$$= f_c + f_b - f_c + f_b$$

$$\therefore \text{Bandwidth} = \underline{\underline{2 f_b}} \quad \text{--- (17)}$$

FREQUENCY SHIFT KEYING [FSK or BFSK]

FSK is defined as shifting the frequency of the carrier wave in accordance with the base-band signal. The amount of frequency shift should be a constant Ω .

therefore modulated signal $s(t)$ is given by

$$s(t) = \begin{cases} \sqrt{2P_s} \cos(2\pi f_c + \Omega)t & s_y = 1 \\ \sqrt{2P_s} \cos(2\pi f_c - \Omega)t & s_y = 0 \end{cases} \quad \text{--- ①}$$

Since binary data are represented by polar NRZ waveform i.e. +ve amplitude for $s_y = 1$ and -ve

amplitude for $s_y = 0$ is

DM-11

$$b(t) = \begin{cases} +1 & \text{for } s_y = 1 \quad 0 \leq t \leq T_b \\ -1 & \text{for } s_y = 0 \quad 0 \leq t \leq T_b \end{cases} \quad \text{--- (2)}$$

The equation (1) can also be written as

$$s(t) = \sqrt{2P_s} \cos[(2\pi f_c + b(t) \Omega) t] \quad \text{--- (3)}$$

Also $S_1(t) = \sqrt{2P_s} \cos 2\pi f_1 t$ for $s_y = 1$ $0 \leq t \leq T_b$

$$S_2(t) = \sqrt{2P_s} \cos 2\pi f_2 t \quad \text{for } s_y = 0 \quad 0 \leq t \leq T_b$$

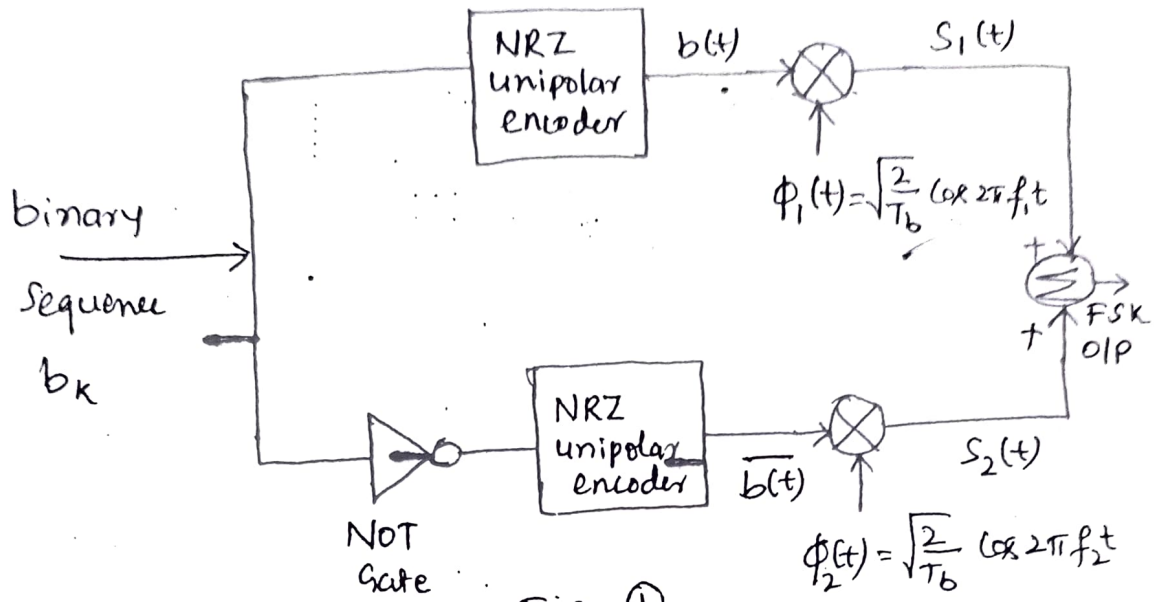
where $f_1 = f_c + \frac{\Omega}{2\pi}$

$$f_2 = f_c - \frac{\Omega}{2\pi}$$

$\Rightarrow$ Generation

The generation of FSK signal is shown in fig - (1). For practical implementation of FSK signal the given data sequence is represented by NRZ unipolar. From the block diagram shown in fig - (1) the data sequence is represented by NRZ unipolar is ASK modulated with a high frequency carrier shown in upper arm of block diagram. The same data sequence is inverted by a NOT gate and resulting sequence is again ASK modulated with low frequency carrier

shown in bottom arm of block diagram fig-①



$$\begin{aligned}
 S_1(t) &= \sqrt{E_b} \sqrt{\frac{2}{T_b}} \cos 2\pi f_1 t \quad \sqrt{E_b} \phi_1(t) \\
 &= \sqrt{\frac{2E_b}{T_b}} \cos 2\pi f_1 t \\
 S_2(t) &= \sqrt{\frac{2E_b}{T_b}} \cos 2\pi f_2 t \quad \sqrt{E_b} \phi_2(t)
 \end{aligned}$$

From block diagram, the signal $b(t)$ and $\bar{b}(t)$ are defined as

$$b(t) = \begin{cases} \sqrt{E_b} & \text{for +ve amplitude (SY-1)} \\ 0 & \text{for '0' amplitude (SY-0)} \end{cases}$$

$$\bar{b}(t) = \begin{cases} 0 & \text{for '0' amplitude (SY-1)} \\ \sqrt{E_b} & \text{for + amplitude (SY-0)} \end{cases}$$

The waveforms for $b(t)$, $\bar{b}(t)$, $S_1(t)$, $S_2(t)$ and $S(t)$ are shown in fig-②

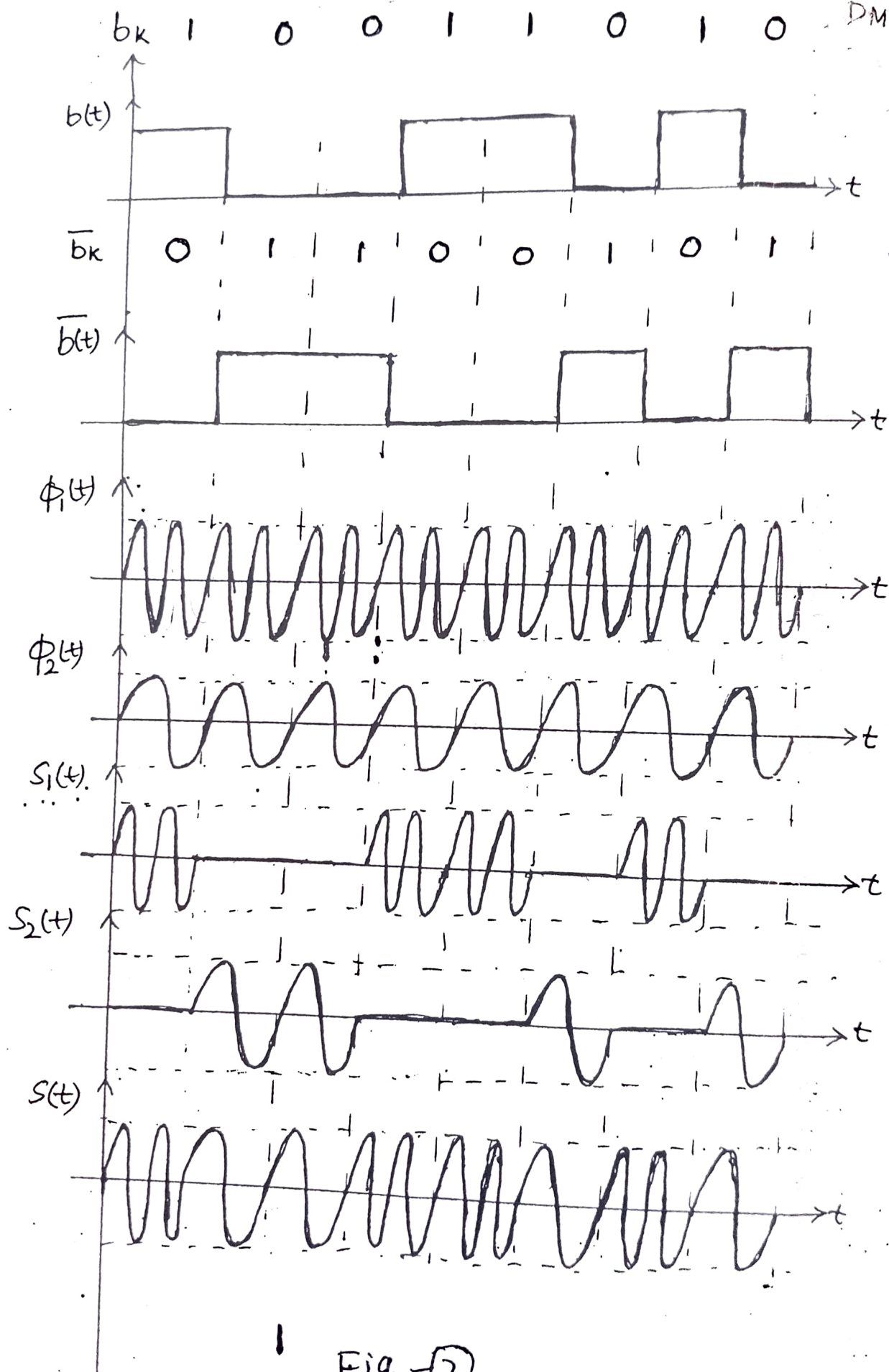


Fig - (2)

⇒ Detection of FSK

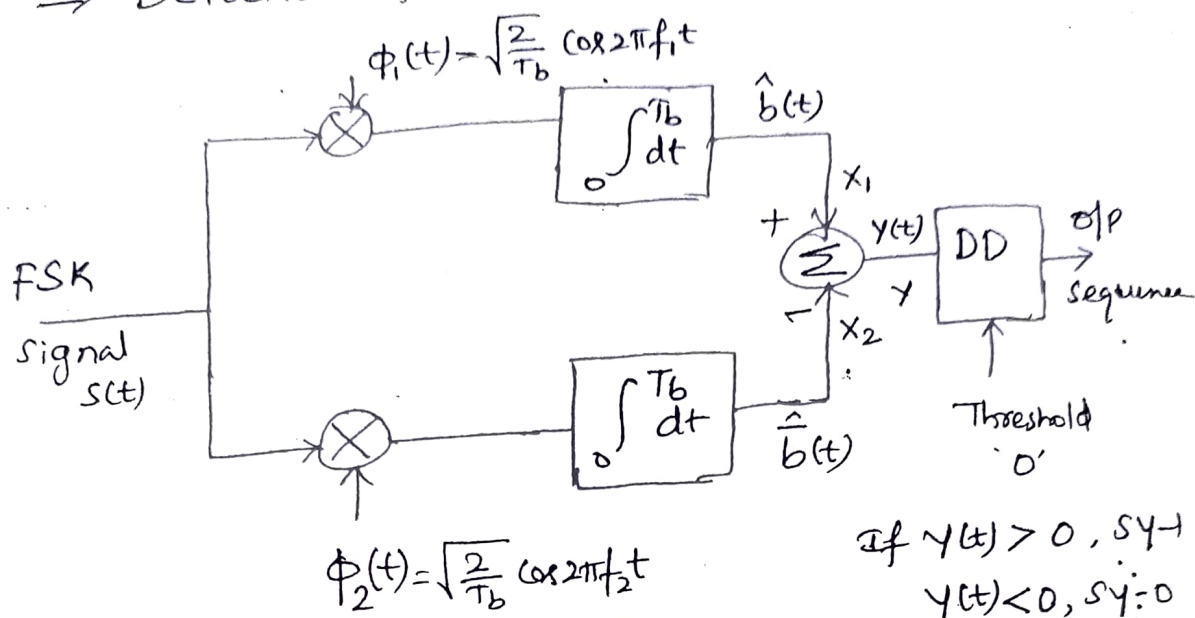


Fig-③

The block diagram of Coherent BFSK receiver is shown in fig-③. The carrier signals $\phi_1(t)$ and $\phi_2(t)$ are locally generated at the receiver and are given to separate multipliers. The locally generated carriers are exactly same as the carrier signals used at the transmitter, hence the fig-③ is called coherent detection of FSK modulated signal. The output of both multipliers are given to integrator, which suppresses the high frequency components and produces the equivalent base band signals are $\hat{b}(t)$ and $\hat{\bar{b}}(t)$. The output of both integrators are subtracted and is given to decision device for comparison and produces the actual base band signal.

For Non-coherent detection of BFSK signal the received signal is pass through separate Band-pass

filters with its center frequencies f_1 and f_2 . DM13
 The output of band-pass filters are applied to separate
 envelope detector, which detects only base band signals
 of $\hat{b}(t)$ and $\hat{b}(t)$ and both signals are compared to get
 the original base band signal $\hat{b}(t)$. The non-coherent
 detection is shown in fig - (4)

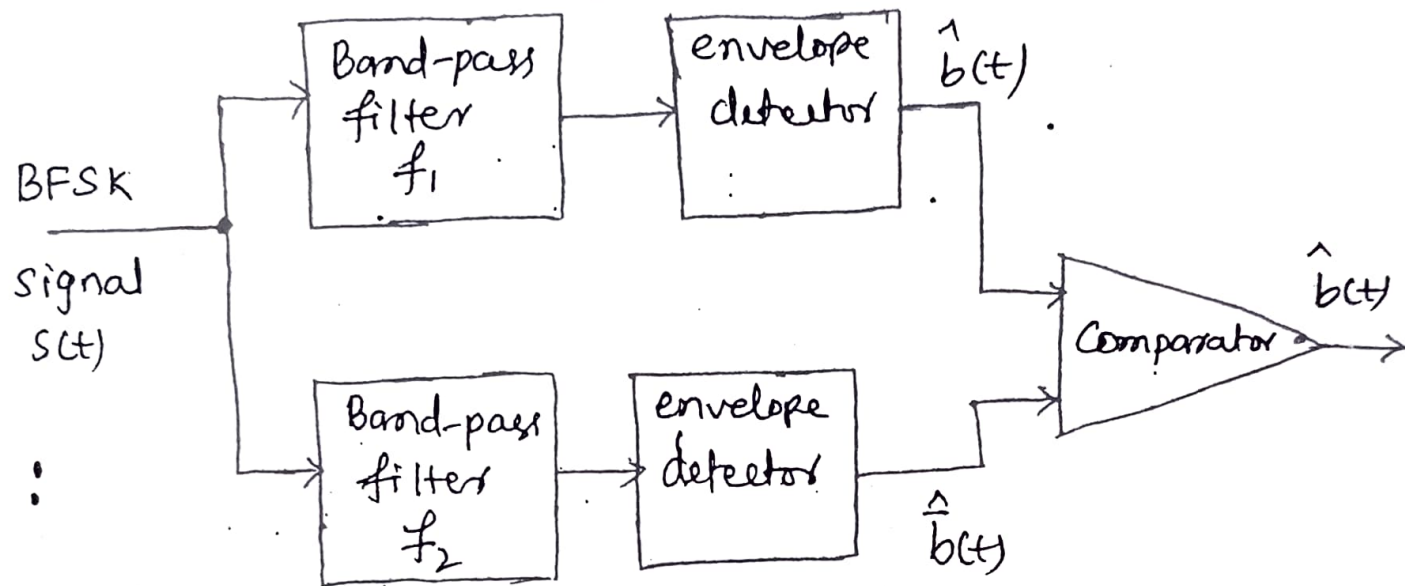


Fig - (4)

⇒ Spectrum of BFSK signal

As we know FSK signal is obtained by adding two ASK signals. Recall the spectrum of ASK signal, in this case the base band signal $b(t)$ is represented in NRZ unipolar and its spectrum $S_B(f)$ is given by

$$S_B(f) = \frac{a^2}{4} \delta(f) + \frac{a^2 T_b}{4} \text{sinc}^2(f T_b) \quad \text{--- ①}$$

In general the spectrum of any band pass signal is given by

$$S_s(f) = \frac{1}{4} [S_B(f - f_c) + S_B(f + f_c)] \quad \text{--- ②}$$

The spectrum of FSK modulated signal is given by

$$S_s(f) = \frac{1}{4} [S_B(f - f_1) + S_B(f + f_1) + S_B(f - f_2) + S_B(f + f_2)] \quad \text{--- ③}$$

Substitute equation ① in ③ i.e.

$$\begin{aligned} S_s(f) = & \frac{a^2}{16} \delta(f - f_1) + \frac{a^2 T_b}{16} \text{sinc}^2[(f - f_1) T_b] \\ & + \frac{a^2}{16} \delta(f + f_1) + \frac{a^2 T_b}{16} \text{sinc}^2[(f + f_1) T_b] \\ & + \frac{a^2}{16} \delta(f - f_2) + \frac{a^2 T_b}{16} \text{sinc}^2[(f - f_2) T_b] \\ & + \frac{a^2}{16} \delta(f + f_2) + \frac{a^2 T_b}{16} \text{sinc}^2[(f + f_2) T_b] \quad \text{--- ④} \end{aligned}$$

The equation ④ represents FSK modulated spectrum is shown in fig-⑥

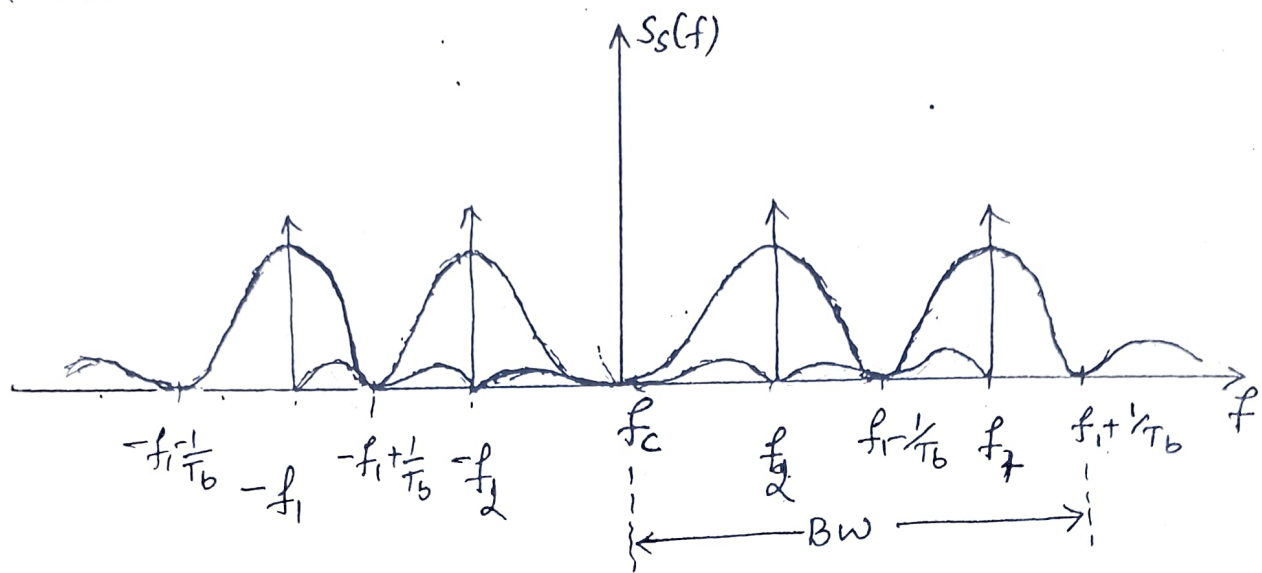


Fig-⑥

⇒ Band width of FSK modulated signal

From spectrum of FSK modulated signal shown in fig⑥ the bandwidth is

$$BW_{FSK} = (f_1 + \frac{1}{T_b}) - (f_2 - \frac{1}{T_b})$$

$$\text{but } f_1 = f_2 + \frac{2}{T_b}$$

$$\therefore BW = (f_2 + \frac{2}{T_b} + \frac{1}{T_b}) - (f_2 - \frac{1}{T_b})$$

$$= f_2 + \frac{3}{T_b} - f_2 + \frac{1}{T_b}$$

$$= \frac{4}{T_b} = \underline{\underline{4f_b}}$$

The FSK modulated signal requires twice the bandwidth required for ASK and PSK modulated signal.

DIFFERENTIAL PHASE SHIFT KEYING [DPSK]

DPSK is a non-coherent detection version of PSK modulation technique. It eliminates the need for a locally generated carrier at the receiver. The DPSK transmitter consists of two basic operations, which are the differential encoder and PSK modulator shown in fig-①. In DPSK modulation technique when symbol '0' is transmitted, then phase of the carrier advances; a phase shift of 180° . Similarly when symbol '1' is transmitted the phase of the carrier is unchanged.

⇒ Generation DPSK Signal

The generation of DPSK signal is shown in fig-①

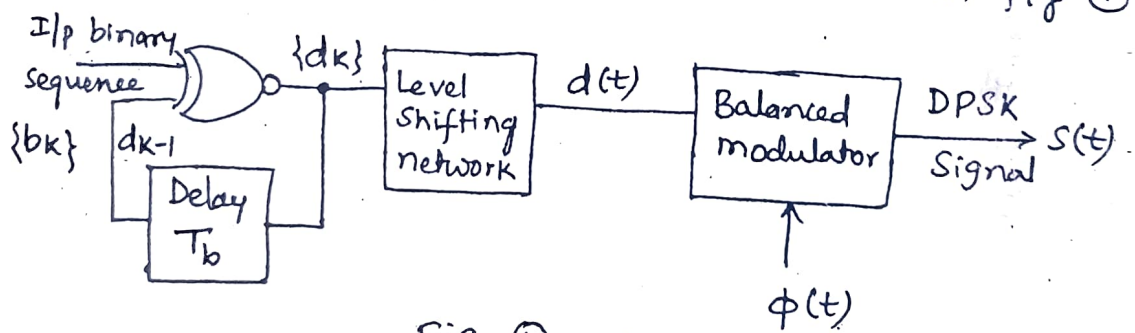


Fig-①

Let $\{b_k\}$ is the arbitrary binary sequence or binary data applied as one input to logic network (XNOR gate). The output of XNOR gate is delayed by one-bit (T_b) interval and applied as the second input to the XNOR gate called $\{d_{k-1}\}$ shown in fig-①. The output of XNOR gate is $\{d_k\}$ is given by

$$d_k = b_k \odot d_{k-1} = \overline{b_k \oplus d_{k-1}}$$

$$= b_k d_{k-1} + \overline{b_k} \overline{d_{k-1}} \quad \text{--- ①}$$

Let us consider an binary sequence $\{b_k\} = \{10010011\}$ as the binary data, to produce the output of XNOR gate when first bit of $\{b_k\}$ is applied by assuming previous output of XNOR gate is at logic high i.e. logic '1'. The DPSK data generated at the output of XNOR for entire $\{b_k\}$ is shown in table-①

$\{b_k\}$	1	0	0	1	0	0	1	1
$\{\overline{b_k}\}$	0	1	1	0	1	1	0	0
d_{k-1}	1	1	0	1	1	0	1	1
$\overline{d_{k-1}}$	0	0	1	0	0	1	0	0
$b_k d_{k-1}$	1	0	0	1	0	0	1	1
$\overline{b_k} \overline{d_{k-1}}$	0	0	1	0	0	1	0	0

DPSK data d_k	1	1	0	1	1	0	1	1
(Assumed bit)								

Transmitting phase sequence (ϕ)	0	0	π	0	0	π	0	0
	1	0	0	1	0	0	1	1

Rx As from fig-①, the output of XNOR gate is not only depends on present input bit from $\{b_k\}$, but it also depends on previous output $\{d_{k-1}\}$ to produce the non-coherent detection is not possible in BPSK. As is done in DPSK, cost is reduced.

output of XNOR Gate. Hence output of differential ^{DM-18} encoder depends on two consecutive bits of $\{b_k\}$, therefore symbol duration 'T' at the output of encoder is $T = 2T_b$.

The output of encoder is given as a baseband signal to the balanced modulator results a DPSK modulated signal $s(t)$ and is given by

$$s(t) = \sqrt{\frac{E_b}{2T_b}} \cos(2\pi f_c t) \quad 0 \leq t \leq T_b \quad \text{--- (2)}$$

When Symbol '1' is transmitted o/p of encoder $s_1(t)$ is given by

$$s_1(t) = \begin{cases} \sqrt{\frac{E_b}{2T_b}} \cos 2\pi f_c t & 0 \leq t \leq T_b \\ \sqrt{\frac{E_b}{2T_b}} \cos 2\pi f_c t & T_b \leq t \leq 2T_b \end{cases} \quad \text{--- (3)}$$

from equation (3), we have phase of the carrier is remains unchanged

When symbol '0' is transmitted, the carrier signal advances a phase shift of 180° . Therefore $s_2(t)$ is given by

$$s_2(t) = \begin{cases} \sqrt{\frac{E_b}{2T_b}} \cos(2\pi f_c t) & 0 \leq t \leq T_b \\ + \sqrt{\frac{E_b}{2T_b}} \cos(2\pi f_c t + \pi) & T_b \leq t \leq 2T_b \end{cases} \quad \text{--- (4)}$$

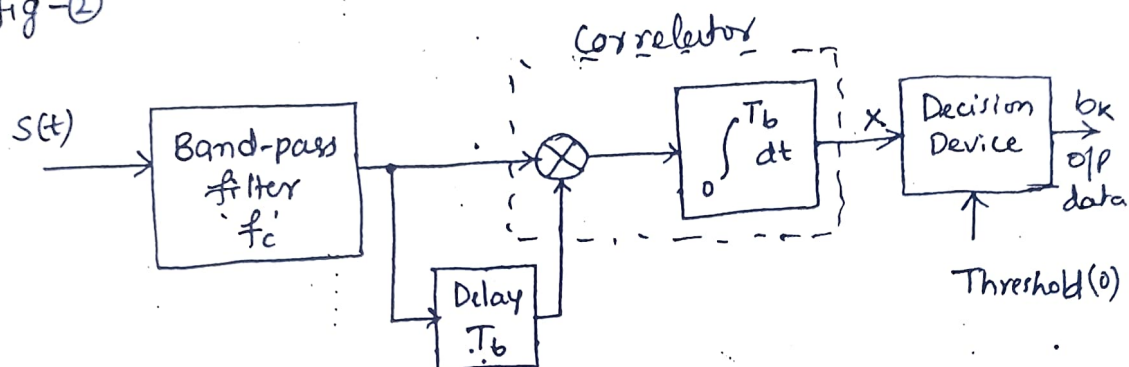
* In DPSK energy of a symbol $E = 2E_b$ and probability error P_e is given by

$$P_e = \frac{1}{2} e^{-E_b/N_0} \quad \text{--- (5)}$$

⇒ Detection DPSK Signal

The detection of DPSK signal is shown in

Fig-2



$$\begin{aligned} \text{when } x > 0 &= s_1 - 1 \\ x < 0 &= s_1 - 0 \end{aligned} \left. \vphantom{\begin{aligned} \text{when } x > 0 \\ x < 0 \end{aligned}} \right\} \text{ of } \{B_k\}$$

The received signal $s(t)$ is passing through a band-pass filter with a center frequency of ' f_c ' in order to limit noise power. The filtered output and delayed version of it is multiplied and multiplied output is given to integrator. Since integrator behaves like low pass filter which gives only base band signal. The resulting base band signal is sampled at $t = T_b$ and its sample ' x ' is compared with the threshold level ' 0 '.

when $x > 0$ The Decision device produces in favour of logic '1' or symbol '1'

when $x < 0$ The decision device produces in favour of logic '0' or symbol '0'

⊗ Assume that, when DPSK modulated signal is transmitted through transmission channel at the receiver the signal gets phase shifted by θ . Therefore received signal is

$$= d(t) \sqrt{\frac{E_b}{2T_b}} \cos(2\pi f_c t + \theta)$$

⊗ The received signal and one bit delayed version of it are given to multiplier, the output of multiplier is

$$= d(t) \sqrt{\frac{E_b}{2T_b}} \cos(2\pi f_c t + \theta) \times d(t - T_b) \cos[2\pi f_c (t - T_b) + \theta]$$

$$= d(t) d(t - T_b) \times \frac{E_b}{2T_b} \cos(2\pi f_c t + \theta) \cos[2\pi f_c (t - T_b) + \theta]$$

$$= d(t) d(t - T_b) \frac{E_b}{2T_b} \times \frac{1}{2} \left[\cos(2\pi f_c t + \theta - 2\pi f_c t + 2\pi f_c T_b - \theta) + \cos(2\pi f_c t + \theta + 2\pi f_c t - 2\pi f_c T_b + \theta) \right]$$

$$= d(t) d(t - T_b) \frac{E_b}{4T_b} \left[\cos 2\pi f_c T_b + \cos(4\pi f_c t - 2\pi f_c T_b + 2\theta) \right]$$

$$= d(t) d(t - T_b) \frac{E_b}{4T_b} \left\{ \cos 2\pi f_c T_b + \cos(4\pi f_c t - 2\pi f_c T_b + 2\theta) \right\}$$

$$= d(t) d(t - T_b) \frac{E_b}{4T_b} \left[\cos 2\pi f_c T_b + \cos \left[4\pi f_c \left(t - \frac{T_b}{2} \right) + 2\theta \right] \right]$$

Let f_c be the carrier signal, T_b is bit duration
 T_b contain 'n' number of cycles of f_c then

$$f_c = n f_b, \quad f_c = \frac{n}{T_b}$$

$$n = f_c T_b$$

Put the value of $n = f_c T_b$ in first term of above equation

$$\cos 2\pi n = 1$$

$\therefore$ o/p of multiplier is

$$= d(t) d(t-T_b) \frac{E_b}{4T_b} + d(t) d(t-T_b) \frac{E_b}{4T_b} \cos \left[4\pi f_c \left(t - \frac{T_b}{2} \right) + 2\theta \right]$$

(*) The above signal is applied to the integrator, then output of it for k^{th} bit interval is given by

$$x(kT_b) = d(kT_b) d[(k-1)T_b] \frac{E_b}{4T_b} \int_{(k-1)T_b}^{kT_b} dt + d(kT_b) d[(k-1)T_b] \frac{E_b}{4T_b} \int_{(k-1)T_b}^{kT_b} \cos \left[4\pi f_c \left(t - \frac{T_b}{2} \right) + 2\theta \right] dt$$

The second term in above equation becomes zero

$$\therefore x(kT_b) = d(kT_b) d[(k-1)T_b] \frac{E_b}{4T_b} [kT_b - \{(k-1)T_b\}]$$

$$= d(kT_b) d[(k-1)T_b] \frac{E_b}{4T_b} [kT_b - kT_b + T_b]$$

$$= d(kT_b) d[(k-1)T_b] \frac{E_b}{4}$$

$$\therefore x(kT_b) = \frac{PT_b}{4} d(kT_b) d[(k-1)T_b] \text{ --- (*)}$$

The above signal is sampled at the rate of $t = T_b$ then sample of it is

$$x = -PT_b \quad d(t) = 0 \quad (\text{-ve amplitude})$$

$$x = PT_b \quad d(t) = 1 \quad (\text{+ve amplitude})$$

$\Rightarrow$ Bandwidth of DPSK

Since symbol duration is $T = 2T_b$

$$\therefore BW = \frac{2}{T} = \frac{2}{2T_b} = \frac{1}{T_b} = \underline{\underline{f_b}}$$

To have reliable digital communication system, the digital modulation technique used should produce low probability of error and requires less bandwidth for transmitting digitally modulated signal. The following digital modulated signals were concentrated on above two parameters. The two techniques are QPSK and MSK modulation techniques

The QPSK is an example of quadrature carrier multiplexing system, which produces the modulated signal in terms of inphase and quadrature phase components

$$S(t) = S_I(t) \cos 2\pi f_c t - S_Q(t) \sin 2\pi f_c t$$

1 QUADRATURE - PHASE SHIFT KEYING [QPSK]

In BPSK, the phase of the carrier is shifted by an amount of 180° for every change in symbols i.e. from 1 to '0' and '0' to 1. But in case of QPSK the phase of the carrier changes by an amount of 90° between the symbols. The QPSK uses phase of the carrier one of four equally spaced values, such as $\pi/4$, $3\pi/4$, $5\pi/4$ and $7\pi/4$. therefore the transmitting phase sequence of QPSK modulated signal is given by

$$S_i(t) = \begin{cases} \sqrt{\frac{2E_s}{T_s}} \cos[2\pi f_c t + (2i-1)\pi/4] & 0 \leq t \leq T_s \\ 0 & \text{otherwise} \end{cases} \quad \text{--- (1)}$$

where $i = 1, 2, 3, 4$

$E_s \Rightarrow$ transmitting signal energy/symbol

$T_s \Rightarrow$ symbol duration

Since in QPSK transmitting two bits per symbol, then signalling symbol rate is decreases, it helps to select the lower bandwidth for signal transmission. In QPSK grouping two data bits per symbol correspondingly we have '4' symbols ($2^2=4$). The symbols and its phase relation is

Symbol	Phase of carrier signal produce in QPSK
10	$\pi/4$
00	$3\pi/4$
01	$5\pi/4$
11	$7\pi/4$

By using a trigonometric function the equation (1) is written as in terms of inphase and quadrature components

$$S_i(t) = \begin{cases} \sqrt{\frac{2E_s}{T_s}} \cos[(2i-1)\pi/4] \cos(2\pi f_c t) - \\ \sqrt{\frac{2E_s}{T_s}} \sin[(2i-1)\pi/4] \sin(2\pi f_c t) & 0 \leq t \leq T_s \\ 0 & \text{otherwise} \end{cases} \quad \text{--- (2)}$$

From equation (2), it is clear that there are two orthogonal carriers are $\phi_1(t)$ and $\phi_2(t)$

$$\left. \begin{aligned} \phi_1(t) &= \sqrt{\frac{2}{T_s}} \cos(2\pi f_c t) & 0 \leq t \leq T_s \\ \phi_2(t) &= \sqrt{\frac{2}{T_s}} \sin(2\pi f_c t) & 0 \leq t \leq T_s \end{aligned} \right\} \text{--- (3)} \quad \text{DM21}$$

Therefore by using equation (2) and (3) the QPSK modulated signal $S_i(t)$ is given by

$$S_i(t) = \sqrt{E_s} \phi_1(t) \cos[(2i-1)\pi/4] - \sqrt{E_s} \phi_2(t) \sin[(2i-1)\pi/4] \text{--- (4)}$$

$\Rightarrow$ QPSK OR OQPSK Transmitter

The block diagram of QPSK transmitter is shown in fig-(1). The digital data or binary sequence generated by digital source are converted into NRZ polar waveform $b(t)$, with symbols '1' and '0' represented by $+\sqrt{E_b}$ and $-\sqrt{E_b}$ Volts respectively. This binary wave $b(t)$ is divided by means of demultiplexer or bit splitter into two binary waves consisting of odd and even numbered input bits. These two binary waves are denoted by $b_o(t)$ and $b_e(t)$ respectively. The $b_o(t)$ and $b_e(t)$ are having a delay of ' T_b ' hence QPSK signal is called Offset-QPSK signal. The signals $b_o(t)$ and $b_e(t)$ are modulated with a pair of carriers signals or orthogonal carriers. These two carriers are having a phase difference of 90° . $b_o(t)$ is modulated with $\sqrt{\frac{2}{T_s}} \cos 2\pi f_c t$ results a PSK modulated signal. Also $b_e(t)$ is modulated with $\sqrt{\frac{2}{T_s}} \sin 2\pi f_c t$ results another PSK modulated signal correspondingly they are called $s_o(t)$ and $s_e(t)$.

$$S_o(t) = b_o(t) \sqrt{\frac{2}{T_s}} \cos(2\pi f_c t) \quad \text{--- (5)}$$

$$S_e(t) = b_e(t) \sqrt{\frac{2}{T_s}} \sin(2\pi f_c t) \quad \text{--- (6)}$$

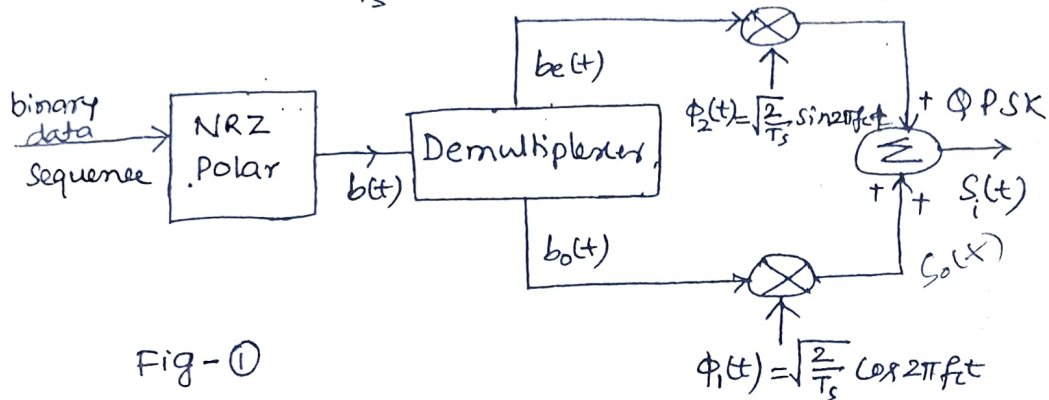


Fig-①

From block diagram shown in fig-① the QPSK modulated signal $s_i(t)$ is given by

$$S_i(t) = S_o(t) + S_e(t) \quad \text{--- (7)}$$

$$\therefore S_i(t) = b_o(t) \sqrt{\frac{2}{T_s}} \cos 2\pi f_c t + b_e(t) \sqrt{\frac{2}{T_s}} \sin 2\pi f_c t \quad \text{--- (8)}$$

where $T_s = 2T_b$ and $E_s = P_s T_s$
 $= P_s \times 2T_b = 2P_s T_b$

Compare equation (4) and (8) we have

$$b_o(t) = b_e(t) = \sqrt{E_s} = \sqrt{2E_b} = \sqrt{2P_s T_b}$$

Let $b_o(t)$ and $b_e(t)$ are given by

$$\therefore b_o(t) = \sqrt{2} \cos[(2i-1)\pi/4] \quad \text{--- (9)}$$

$$b_e(t) = -\sqrt{2} \sin[(2i-1)\pi/4] \quad \text{--- (10)}$$

Therefore QPSK modulated signal $S_i(t)$ is given by

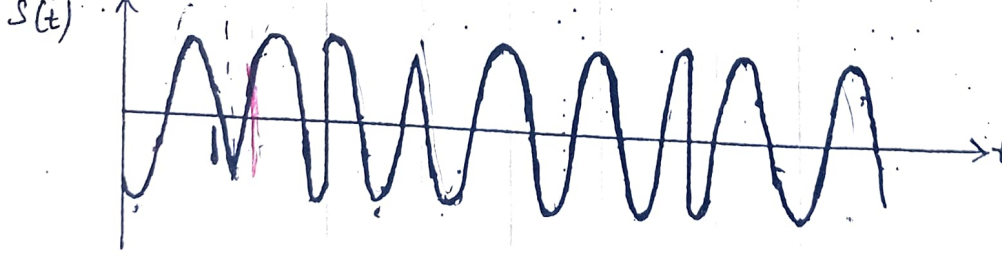
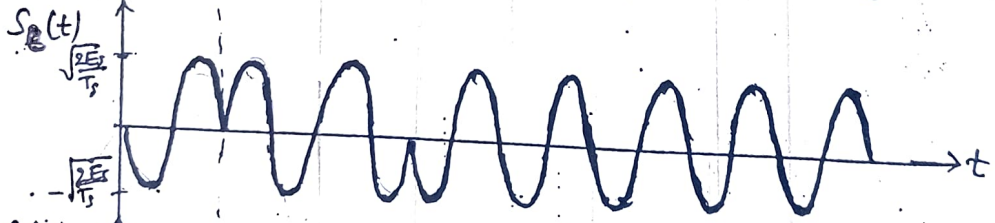
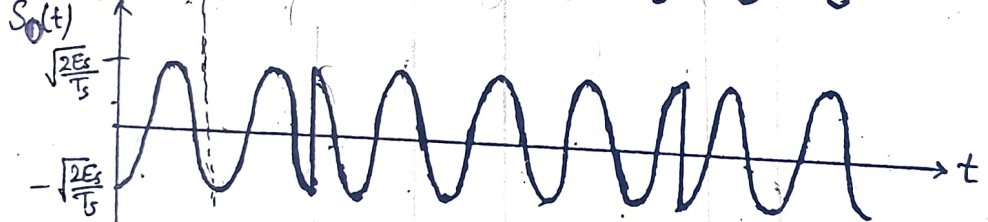
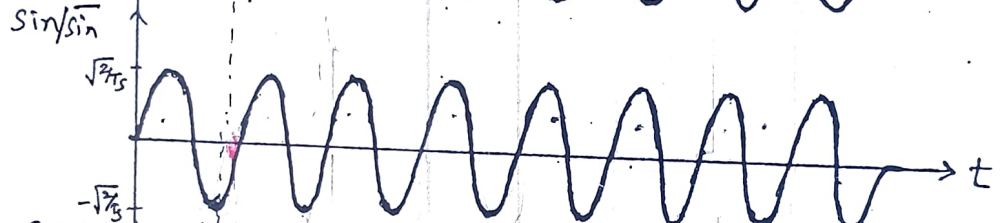
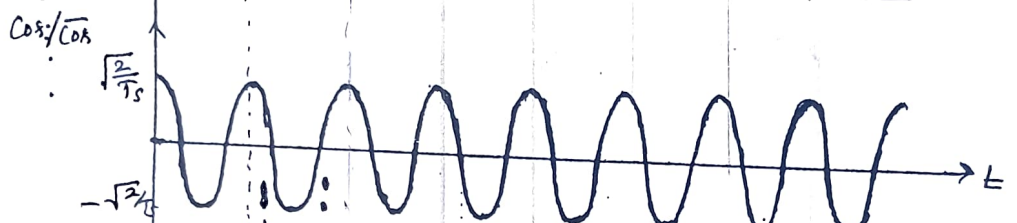
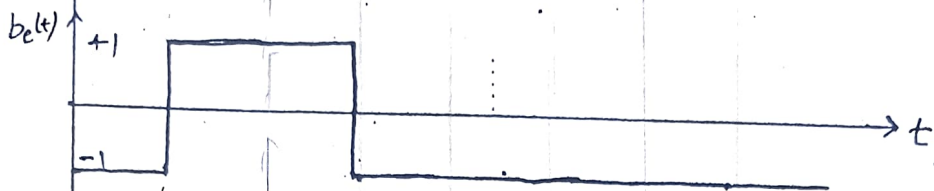
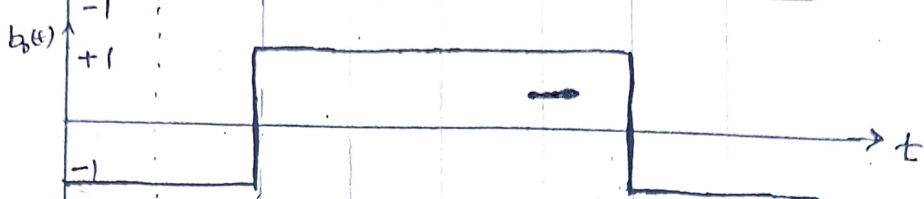
$$S_i(t) = \sqrt{P_s T_b} b_o(t) \phi_1(t) + \sqrt{P_s T_b} b_e(t) \phi_2(t) \quad \text{--- (11)}$$

$$= \sqrt{E_b} b_o(t) \phi_1(t) + \sqrt{E_b} b_e(t) \phi_2(t) \quad \text{--- (12)}$$

where $E_b = P_s T_b$

binary
Sequence

0 1 1 0 1 0 0 0 0 0 DM22



$\rightarrow 45^\circ, 135^\circ, 225^\circ, 315^\circ$

⇒ QPSK Detection

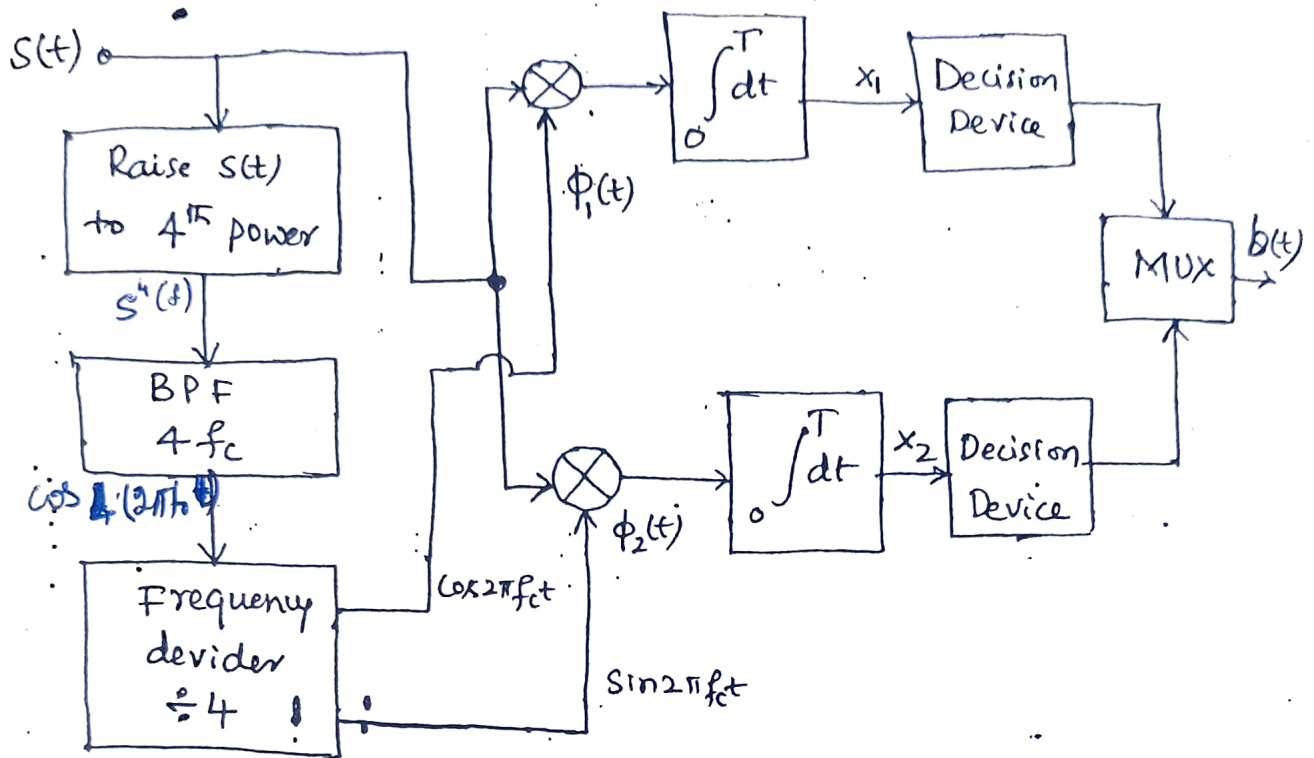


Fig - (3)

The QPSK detection is shown in fig-(3). The received signal $S(t)$ is raised to 4^{th} power, then passing signal through a band-pass filter with a center frequency $4f_c$. At the output of band-pass filter the frequency of the carrier wave has been increased to FOUR times the actual frequency, hence output of it is pass through a Frequency divider ($\div 4$) to have same frequency of the carrier. Therefore the output of frequency divider produces an locally generated orthogonal carriers, which are exactly same as that of carriers used at the transmitter side they are $\phi_1(t)$ and $\phi_2(t)$. These locally generated carriers are

used to detect baseband signal coherently. Again, ^{DMZ} the received signal is multiplied with locally generated carriers $\phi_1(t)$ and $\phi_2(t)$ separately shown in fig-(3). The multiplied output is given to integrators of one symbol duration. The output of integrator are sampled over one symbol duration and we get samples as x_1 and x_2 . These samples are compared with threshold level of individual decision device and finally both the outputs of decision device are multiplexed to get the desired baseband signal.

⇒ Spectrum of QPSK signal

The spectrum of QPSK modulated signal is similar to PSK modulated signal spectrum only difference is that, there are only two symbols in PSK and duration of symbol is ' T_b ' (one bit/symbol) but in QPSK, there are four symbols with a symbol duration $T_s = 2T_b$ (Two bits/symbol). Hence T_b in PSK spectrum is replaced by $2T_b$ for QPSK spectrum.

⇒ Bandwidth of QPSK signal

$$\begin{aligned}
 BW &= \left[f_c + \frac{1}{T_s} \right] - \left[f_c - \frac{1}{T_s} \right] \\
 &= f_c + \frac{1}{T_s} - f_c + \frac{1}{T_s} \\
 &= \frac{2}{T_s} = \frac{2}{2T_b} = \frac{1}{T_b} = f_b
 \end{aligned}$$

(The bandwidth of QPSK required is half of the bandwidth required for PSK.)

2. MINIMUM SHIFT KEYING (MSK)

As we studied QPSK modulation technique in which a 90° phase shift produces between symbols. This creates an abrupt amplitude variations in the modulated waveform. Therefore bandwidth of QPSK requires more. Some filters can be used to overcome amplitude variations present in QPSK modulated wave but they may have some other side effects. Hence MSK is used to overcome these effects. In MSK the output waveform is continuous in its phase, hence there are no abrupt change in amplitude. The side lobes of MSK are very small, hence bandpass filtering is not required to avoid interchannel interference. The MSK modulation is obtained similar to QPSK is shown in fig-①

From fig-①, the duration of $b_0(t)$ and $b_1(t)$ is $2T_b$. The two waveforms are generated, which are $\sin 2\pi(t/4T_b)$ and $\cos 2\pi(t/4T_b)$, and they are multiplied by $b_1(t)$ and $b_0(t)$ respectively shown in fig-1(e) & (f). The MSK modulated signal $s(t)$ is given by

$$s(t) = \sqrt{2P_s} \left[b_1(t) \sin(2\pi t/4T_b) \right] \cos 2\pi f_c t + \sqrt{2P_s} \left[b_0(t) \cos(2\pi t/4T_b) \right] \sin 2\pi f_c t \quad \text{--- ①}$$

The equation ① is obtained by $b_1(t) \sin(2\pi t/4T_b)$ and $b_0(t) \cos(2\pi t/4T_b)$ are modulated with quadrature carriers.

MSK is special case of continuous phase FSK. No abrupt change in o/p waveform.

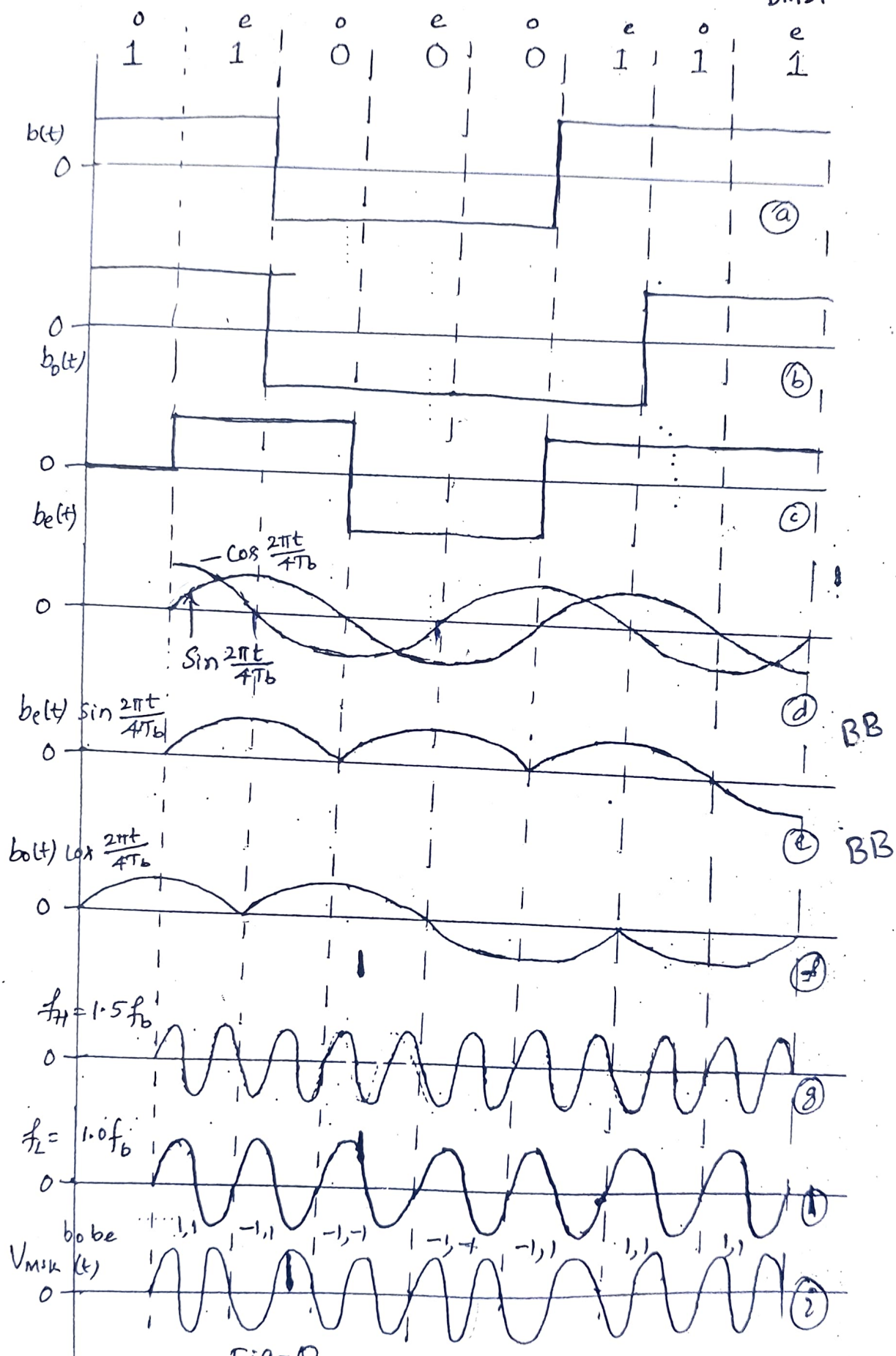


Fig-10

Same phase = high freq
 different phase = low freq

$$s(t) = \sqrt{2P_s} [b_o(t) \times \frac{1}{2} \left[\sin(2\pi(f_c + \frac{1}{4T_b})t) + \sin(2\pi(f_c - \frac{1}{4T_b})t) \right] + \sqrt{2P_s} [b_o(t) \times \frac{1}{2} \left[\sin(2\pi(f_c + \frac{1}{4T_b})t) - \sin(2\pi(f_c - \frac{1}{4T_b})t) \right]]$$

$$2 \sin(x) \cos(y) = \sin(x-y) + \sin(x+y)$$

$$S(t) = \sqrt{2P_s} \left[\frac{b_o(t) + b_e(t)}{2} \right] \sin 2\pi(f_c + \frac{1}{4T_b})t$$

$$+ \sqrt{2P_s} \left[\frac{b_o(t) - b_e(t)}{2} \right] \sin 2\pi(f_c - \frac{1}{4T_b})t$$

$$S(t) = \sqrt{2P_s} \left[\frac{b_o(t) + b_e(t)}{2} \right] \sin 2\pi(f_c + \frac{f_b}{4})t$$

$$+ \sqrt{2P_s} \left[\frac{b_o(t) - b_e(t)}{2} \right] \sin 2\pi(f_c - \frac{f_b}{4})t \quad \text{--- (2)}$$

$$\text{Let } C_H(t) = \frac{b_o(t) + b_e(t)}{2} \text{ \& } C_L(t) = \frac{b_o(t) - b_e(t)}{2} \quad \text{--- (3)}$$

$$f_H = f_c + \frac{f_b}{4} \text{ \& } f_L = f_c - \frac{f_b}{4} \quad \text{--- (4)}$$

By using equation (3) and (4), the equation (2) is :

$$S(t) = \sqrt{2P_s} C_H(t) \sin 2\pi f_H t + \sqrt{2P_s} C_L(t) \sin 2\pi f_L t \quad \text{--- (5)}$$

when $b_o(t) = b_e(t)$, $C_L(t) = 0$ \& $C_H(t) = \pm 1$

$$\therefore S(t) = \sqrt{2P_s} C_H(t) \sin 2\pi f_H t \quad \text{--- (6)}$$

when $b_o(t) = -b_e(t)$, $C_H(t) = 0$ \& $C_L(t) = \pm 1$

$$\therefore S(t) = \sqrt{2P_s} C_L(t) \sin 2\pi f_L t \quad \text{--- (7)}$$

equation (6) and (7) represents only f_H and f_L are being transmitted for that interval.

The frequency ' f_H ' and ' f_L ' are actually taken as orthogonal carriers i.e. $\cos(2\pi f_H t)$ and $\sin(2\pi f_L t)$. For orthogonality following relation should be satisfied

$$\int_0^{T_b} \sin(2\pi f_H t) \sin(2\pi f_L t) dt = 0 \quad \text{--- (8)}$$

The equation (8) is satisfied, if we have integers 'm' and 'n' such that

$$\left. \begin{aligned} 2\pi(f_H - f_L)T_b &= n\pi \\ 2\pi(f_H + f_L)T_b &= m\pi \end{aligned} \right\} \text{--- (9)}$$

Substitute f_H & f_L in equation (9) we have

$$\begin{aligned} 2\pi\left(f_L + \frac{f_b}{4} - f_L + \frac{f_b}{4}\right)T_b &= n\pi \\ \left(\frac{4f_b}{4}\right)T_b &= n \quad \therefore n=1 \text{ --- (10)} \end{aligned}$$

Similarly

$$\begin{aligned} 2\pi\left(f_L + \frac{f_b}{4} + f_L - \frac{f_b}{4}\right)T_b &= m\pi \\ 4f_L T_b &= m, \quad m = 4f_L \times \frac{1}{f_b} \\ \therefore f_L &= \frac{mf_b}{4} \text{ --- (11)} \end{aligned}$$

with $n=1$, equation (9) is given by

$$2\pi(f_H - f_L)T_b = 1 \times \pi$$

$$f_H - f_L = \frac{1}{2T_b}$$

$$\therefore f_H - f_L = \frac{f_b}{2} \text{ --- (12)}$$

For $n=1$, the difference between f_H and f_L is minimum and at the same time they are orthogonal. Hence this technique is called minimum shift keying.

As we know $f_H = f_c + \frac{f_b}{4}$ & $f_L = f_c - \frac{f_b}{4}$

from equation (11) f_H and f_L are given by

$$\begin{aligned} f_H &= \frac{mf_b}{4} + \frac{f_b}{4} = mf \\ &= (m+1) \frac{f_b}{4} \quad \text{--- (13)} \end{aligned}$$

$$\begin{aligned} f_L &= \frac{mf_b}{4} - \frac{f_b}{4} = \\ &= (m-1) \frac{f_b}{4} \quad \text{--- (14)} \end{aligned}$$

When $m=5$

$$f_H = (5+1) \frac{f_b}{4} = \frac{6}{4} f_b = 1.5 f_b$$

$$f_L = (5-1) \frac{f_b}{4} = 1.0 f_b$$

The above calculation shows the number of cycles per bit period i.e. for $\sin 2\pi f_H t$ $1\frac{1}{2}$ cycles and for $\sin 2\pi f_L t$ 1 cycle shown in fig-1 (g) & (h). The signal $s(t)$ of MSK is basically a FSK with reduced bandwidths and produces continuous phase.

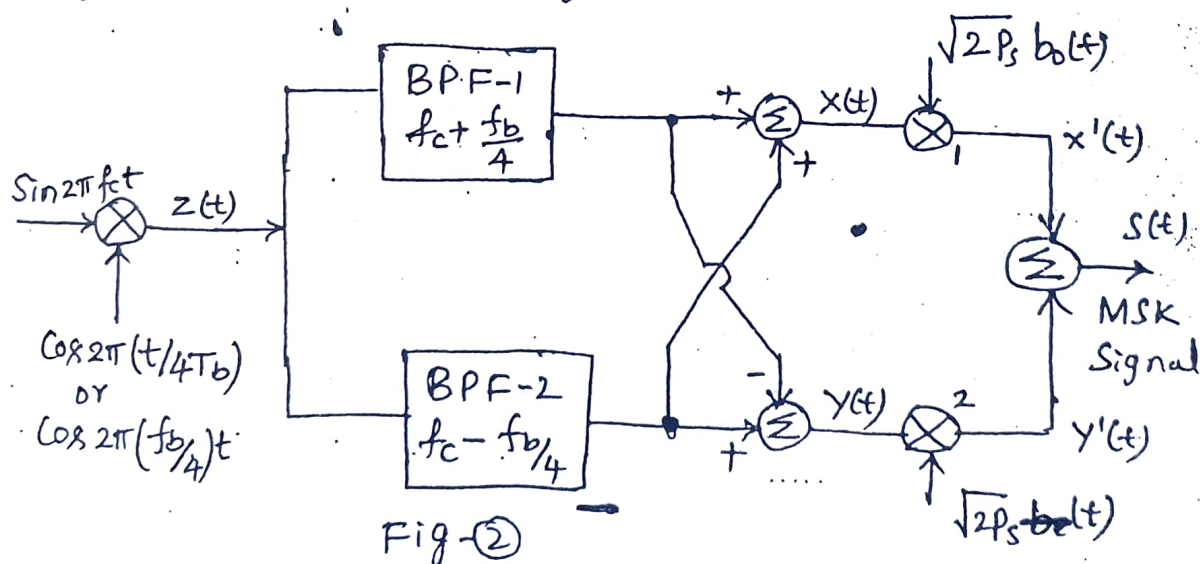
$\Rightarrow$ MSK is shaped QPSK

In QPSK, the $b_e(t)$ and $b_o(t)$ are directly multiply with carrier, hence there are abrupt changes in phase and amplitude. In MSK the two waveforms $b_e(t) \sin(2\pi t/4T_b)$ and $b_o(t) \cos(2\pi t/4T_b)$ are first generated

then these waveforms are multiplied with the carriers.^{upper}
 Thus the multiplied waveform does not have abrupt phase change instead the phase is continuous with respect to time hence MSK is called a shaped QPSK.

⇒ MSK Transmitter

The block diagram of MSK transmitter is shown in fig-②. The two sinusoidal signals are multiplied are $\sin(2\pi f_c t)$ and $\cos(2\pi t/4T_b)$. The multiplied signal is pass through a band pass filters with a center frequency of $f_c + f_b/4$ and $f_c - f_b/4$ are shown in fig-②. The output of band pass filters are added and subtracted resulting signals are identified as $x(t)$ and $y(t)$. Again $x(t)$ is multiplied by $\sqrt{2P_s} b_o(t)$ and $y(t)$ is multiplied with $\sqrt{2P_s} b_e(t)$, then finally adding these two signals to get MSK modulated signal.



$$Z(t) = \sin 2\pi f_c t * \cos 2\pi t (f_b/4) \quad \text{--- ①}$$

$$= \frac{1}{2} [\sin 2\pi (f_c - f_b/4) t + \sin 2\pi (f_c + f_b/4) t] \quad \text{--- ②}$$

The output of 1st BPF is given by

$$\frac{1}{2} \sin 2\pi (f_c + f_b/4) t \quad \text{--- (3)}$$

the output of 2nd BPF is given by

$$\frac{1}{2} \sin 2\pi (f_c - f_b/4) t \quad \text{--- (4)}$$

$$\therefore x(t) = \frac{1}{2} \sin 2\pi (f_c + f_b/4) t + \frac{1}{2} \sin 2\pi (f_c - f_b/4) t$$

$$x(t) = \sin 2\pi f_c t \cos(2\pi t/4T_b) \quad \text{--- (5)}$$

$$\therefore y(t) = \frac{1}{2} \sin 2\pi (f_c - f_b/4) t - \frac{1}{2} \sin 2\pi (f_c + f_b/4) t$$

$$= \frac{1}{2} \left[\sin 2\pi (f_c - f_b/4) t - \sin 2\pi (f_c + f_b/4) t \right]$$

$$= \sin(2\pi t/4T_b) \cos 2\pi f_c t \quad \text{--- (6)}$$

Therefore equation (5) and (6) are again multiplied by $\sqrt{2P_s} b_o(t)$ and $\sqrt{2P_s} b_e(t)$ then

$$x'(t) = \sqrt{2P_s} b_o(t) \sin 2\pi f_c t \cos(2\pi t/4T_b) \quad \text{--- (7)}$$

$$y'(t) = \sqrt{2P_s} b_e(t) \sin(2\pi t/4T_b) \cos 2\pi f_c t \quad \text{--- (8)}$$

Finally MSK modulated signal is given by

$$S(t) = \sqrt{2P_s} \left[b_e(t) \sin(2\pi t/4T_b) \right] \cos 2\pi f_c t + \sqrt{2P_s} \left[b_o(t) \cos(2\pi t/4T_b) \right] \sin 2\pi f_c t \quad \text{--- (9)}$$

$\Rightarrow$ MSK Receiver

The block diagram of coherent detection of MSK signal is shown in fig-(3). The received signal $s(t)$ is multiplied with $x(t)$ and $y(t)$, the output of multiplier

is applied to an integrator and its output is sampled for every bit interval. The sample is compared in a decision device, finally we get the base band signal $b(t)$.

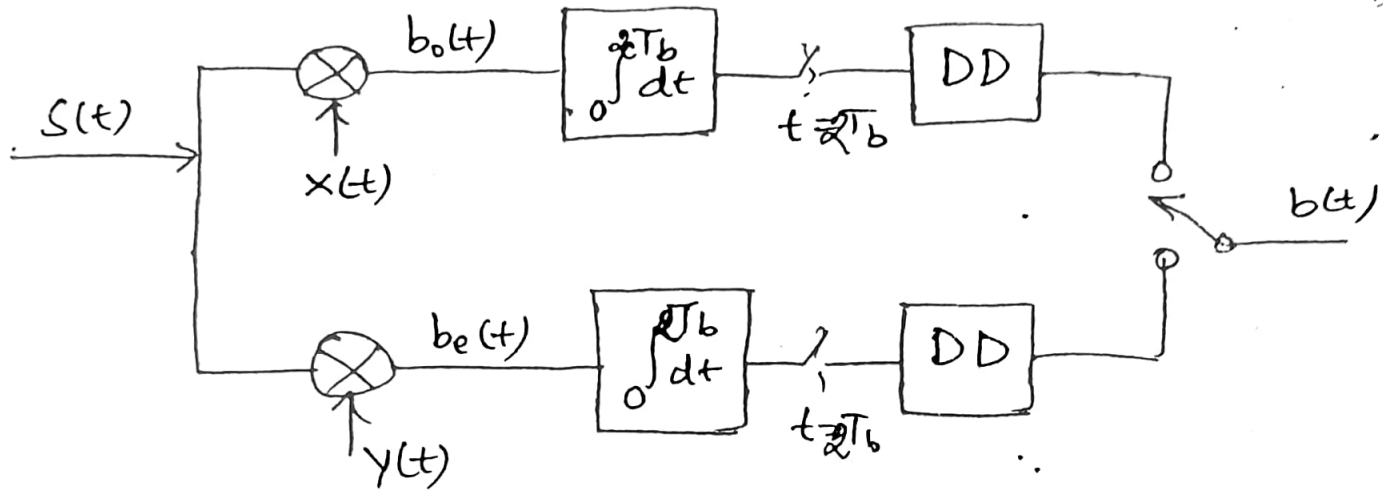


Fig-③

⇒ Power spectral density of MSK

Consider the actual base band signal is given by

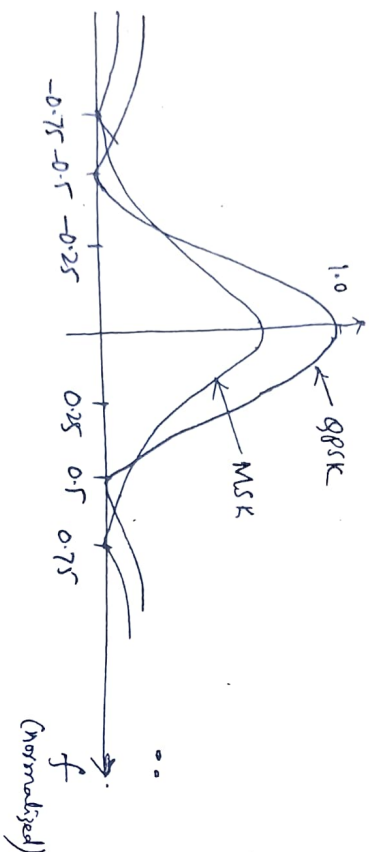
$$B(t) = \sqrt{2P_s} \left[b_o(t) \cos\left(\frac{2\pi t}{4T_b}\right) \right] \\ = \sqrt{2P_s} b_o(t) \cos\left(\pi f_b t / 2\right) \quad \text{--- (1)}$$

The PSD of above equation is given by

$$S_B(f) = \frac{32E_b}{\pi^2} \left[\frac{\cos\left(\frac{2\pi f T_b}{2}\right)}{1 - (4f T_b)^2} \right]^2 \quad \text{--- (2)}$$

When above signal modulated with f_c then

$$S(f) = \frac{8E_b}{\pi^2} \left\{ \frac{\cos 2\pi(f - f_c)T_b}{1 - [4(f - f_c)T_b]^2} \right\}^2 + \frac{8E_b}{\pi^2} \left\{ \frac{\cos 2\pi(f + f_c)T_b}{1 - [4(f + f_c)T_b]^2} \right\}^2$$



⇒ BW of MSK

$$BW = 0.75f_b - (-0.75f_b) \\ = 1.5f_b$$

Bandwidth of MSK higher than QPSK

⇒ Phase continuity in MSK

Let us consider MSK modulated signal $s(t)$ is

$$s(t) = \sqrt{2P_s} \left[\frac{b_o(t) + b_e(t)}{2} \right] \sin 2\pi \left(f_c + \frac{f_b}{4} \right) t \\ + \sqrt{2P_s} \left[\frac{b_o(t) - b_e(t)}{2} \right] \sin 2\pi \left(f_c - \frac{f_b}{4} \right) t \quad \text{--- (1)}$$

$b_e(t)$	$b_o(t)$	$s(t) / \sqrt{2P_s}$
00	-1	$-\sin 2\pi \left(f_c + \frac{f_b}{4} \right) t$
01	-1	$\sin 2\pi \left(f_c - \frac{f_b}{4} \right) t$
10	-1	$-\sin 2\pi \left(f_c - \frac{f_b}{4} \right) t$
11	1	$\sin 2\pi \left(f_c + \frac{f_b}{4} \right) t$

Using above table, the signal $s(t)$ alternatively given by

$$s(t) = b_o(t) \sqrt{2P_s} \sin 2\pi \left[f_c + \frac{b_e(t) f_b}{4} \right] t \quad \text{--- (2)}$$

In equation (2) $b_o(t)$ decides the sign of MSK signal $s(t)$ and $b_e(t) b_o(t)$ decides the frequency. Therefore equation (2) is

$$s(t) = b_o(t) \sqrt{2P_s} \sin \phi(t) \quad \text{--- (3)}$$

$$\text{where } \phi(t) = 2\pi \left[f_c + b_o(t) \frac{b_e(t) f_b}{4} \right] t \quad \text{--- (4)}$$

when $b_e(t)b_o(t) = +1$

$$\phi(t) = \phi_+(t) = 2\pi \left[f_c + \frac{f_b}{4} \right] t \quad \text{--- (5)}$$

when $b_e(t)b_o(t) = -1$

$$\phi(t) = \phi_-(t) = 2\pi \left[f_c - \frac{f_b}{4} \right] t \quad \text{--- (6)}$$

Since $b_e(t)$ and $b_o(t)$ are staggered by T_b , therefore the product $b_e(t) \cdot b_o(t)$ can change at integer multiples of T_b i.e. kT_b , equation (5) & (6) are

$$\phi_+(t) = 2\pi f_c t + \frac{\pi t}{2T_b}$$

$$\phi_-(t) = 2\pi f_c t - \frac{\pi t}{2T_b}$$

The phase change can be represented by difference of above two equations

$$\phi_{diff}(t) = \phi_+(t) - \phi_-(t) = \frac{\pi t}{T_b} \quad \text{--- (7)}$$

As we know that $b_e(t)$ will change its sign at even bits i.e. $t = 2T_b, 4T_b, 6T_b$ etc

$$\phi_{diff}(t) = \frac{\pi t}{T_b} \quad \text{at } t = 2T_b, 4T_b, \dots$$

$$\phi_{diff}(t) = 2\pi, 4\pi, 6\pi, \dots \quad \text{--- (8)}$$

Similarly $b_o(t)$ will change its sign only at odd bit times i.e. when $t = T_b, 3T_b, 5T_b, \dots$

$$\phi_{diff}(t) = \frac{\pi t}{T_b} \quad \text{at } t = T_b, 3T_b, 5T_b, \dots$$

$$\phi_{diff}(t) = \pi, 3\pi, 5\pi, \dots \quad \text{--- (9)}$$

The equation (8) & (9) shows there is no phase shift in MSK signal.

Ex-1 A 4-ary PSK has the transmitted waveforms

$$s_i(t) = 4 \cos(2\pi f_c t + \frac{i\pi}{2}), \quad i = 0, 1, 2, 3 \quad 0 \leq t \leq T$$

Draw its signal space diagram.

Soln :- The M-ary PSK signal is represented by

$$s(t) = \sqrt{2P_s} \cos(2\pi f_c t + \phi_m)$$

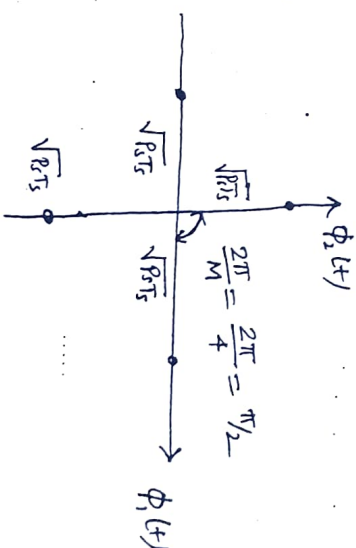
$$\phi_m = (2m+1)\pi/M$$

$$\therefore \sqrt{2P_s} = 4, \quad 2P_s = 16, \quad P_s = \frac{16}{2} = 8$$

$$\phi_0 = \phi_m = \frac{i\pi}{2}, \quad i = 0, 1, 2, 3$$

$$\phi_0 = 0, \quad \phi_1 = \pi/2, \quad \phi_2 = \pi, \quad \phi_3 = 3\pi/2$$

The signal points on signal space diagram of 4-ary PSK is shown below. The signal points are placed at $\sqrt{E_s} = \sqrt{P_s T}$



COMPARISON OF BINARY AND QUATERNARY MODULATION TECHNIQUES

The two systems having unequal number of symbols are being compared.

1. The error rates for all the systems decrease monotonically with increasing values of E_b/N_0

Coherent modulation technique

error probability P_e

$$P_e = \frac{1}{2} \operatorname{erfc}(\sqrt{E_b/N_0})$$

$$\operatorname{erfc}(\sqrt{E_b/N_0}) - \frac{1}{2} \operatorname{erfc}^2(\sqrt{E_b/N_0})$$

$$\frac{1}{2} \operatorname{erfc}(\sqrt{E_b/2N_0})$$

$$\frac{1}{2} e^{-E_b/N_0}$$

$$\frac{1}{2} e^{-E_b/2N_0}$$

$$\operatorname{erfc}(\sqrt{E_b/N_0})$$

$$\operatorname{erfc}(\sqrt{E_b/N_0}) - \frac{1}{4} \operatorname{erfc}^2(\sqrt{E_b/N_0})$$

2. For any given value of E_b/N_0 , coherent PSK produces a smaller error rate than any of the other systems

3. The coherent PSK and DPSK requires an E_b/N_0 of 3dB less when compared to FSK to have the same error rate

4. At high values of E_b/N_0 , DPSK and non coherent FSK perform almost as well as coherent PSK and

conventional coherent FSK, respectively for the same bit rate and signal energy per bit

5. In QPSK two orthogonal carriers $\sqrt{2} \sin \pi f_c t$ and $\sqrt{2} \cos \pi f_c t$ are used, where carrier frequency f_c is an integral multiple of the symbol rate $1/T_s$. At high values of E_b/N_0 , coherent detection of PSK and QPSK have almost same error rate performance

6. MSK and QPSK have almost same error rate performance